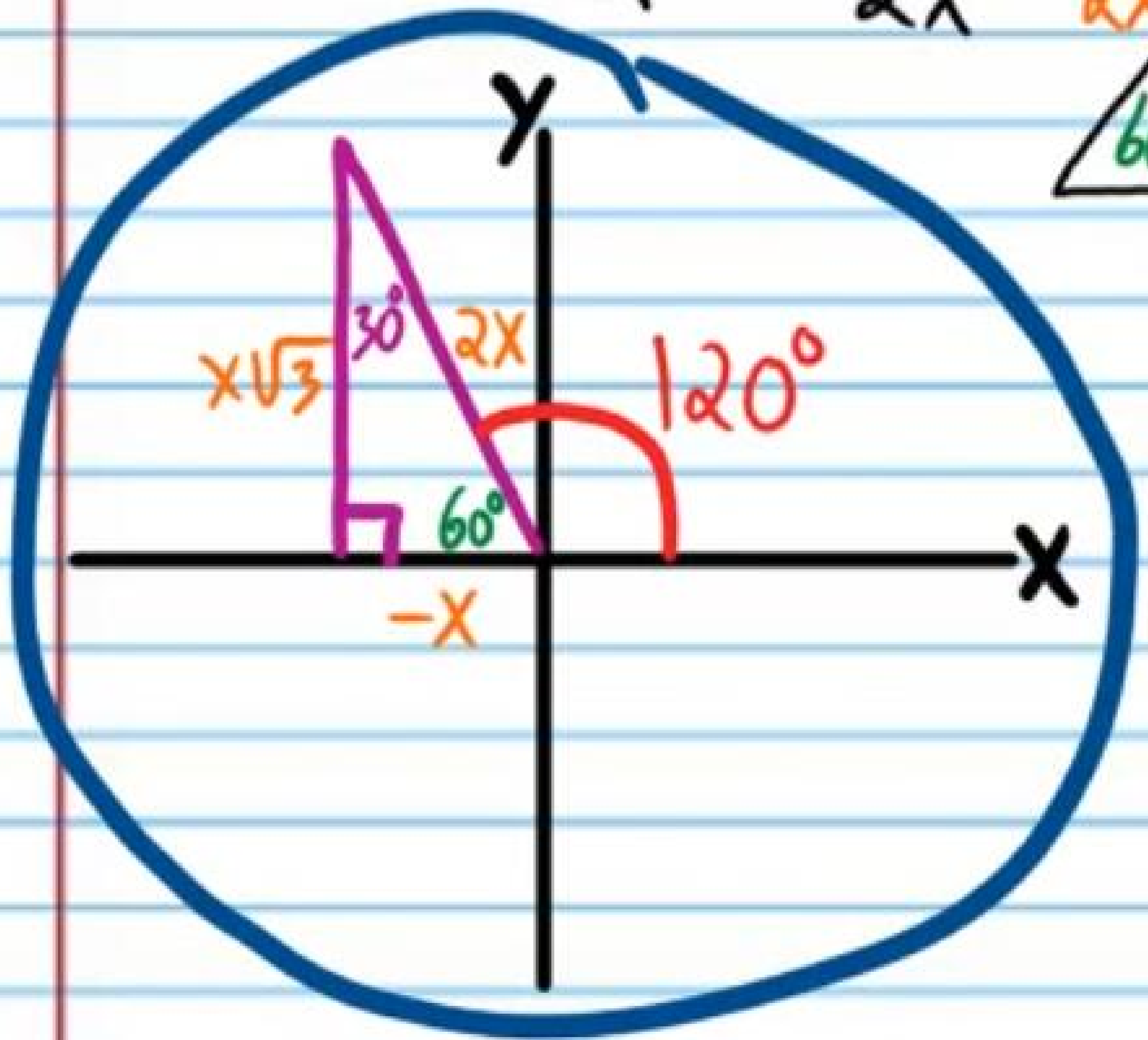
Name
Date
Course

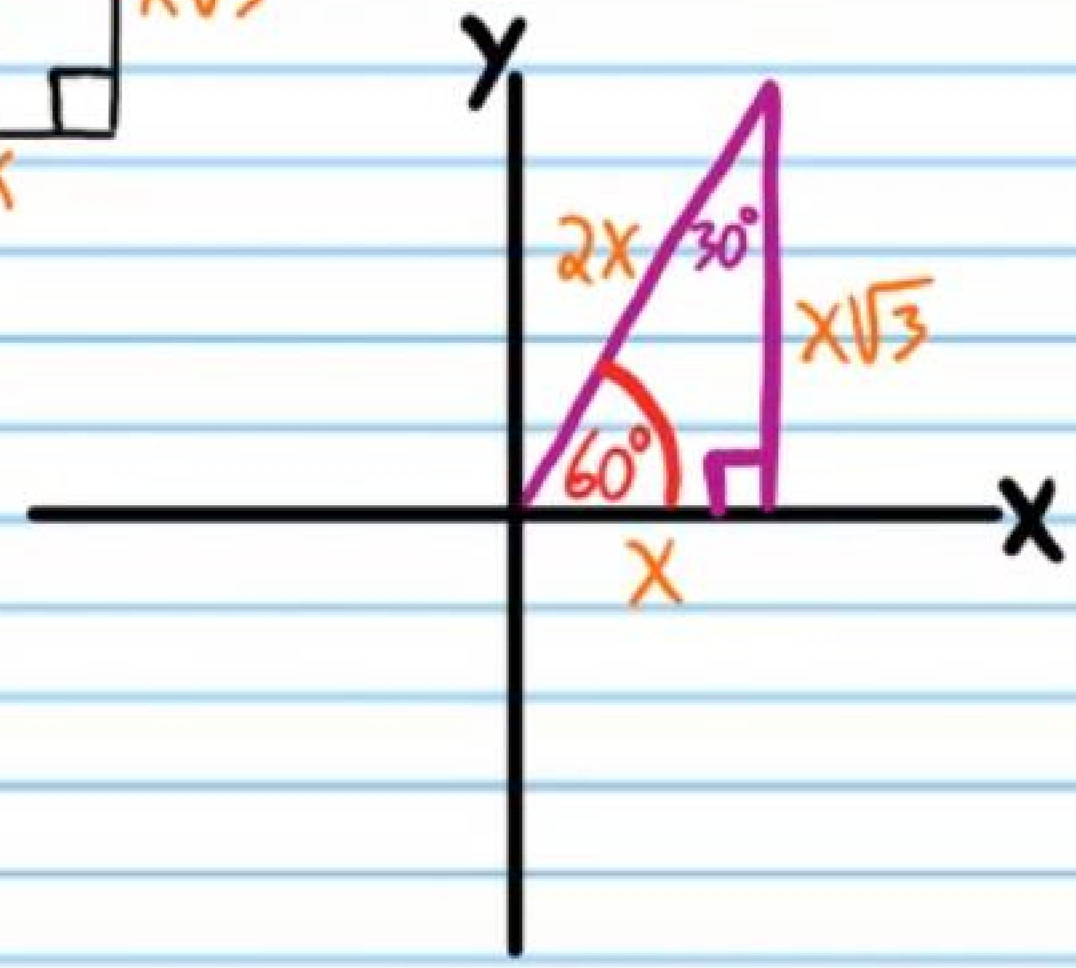
Continued Exploration of the Six Basic Trigonometric Functions (Part II)

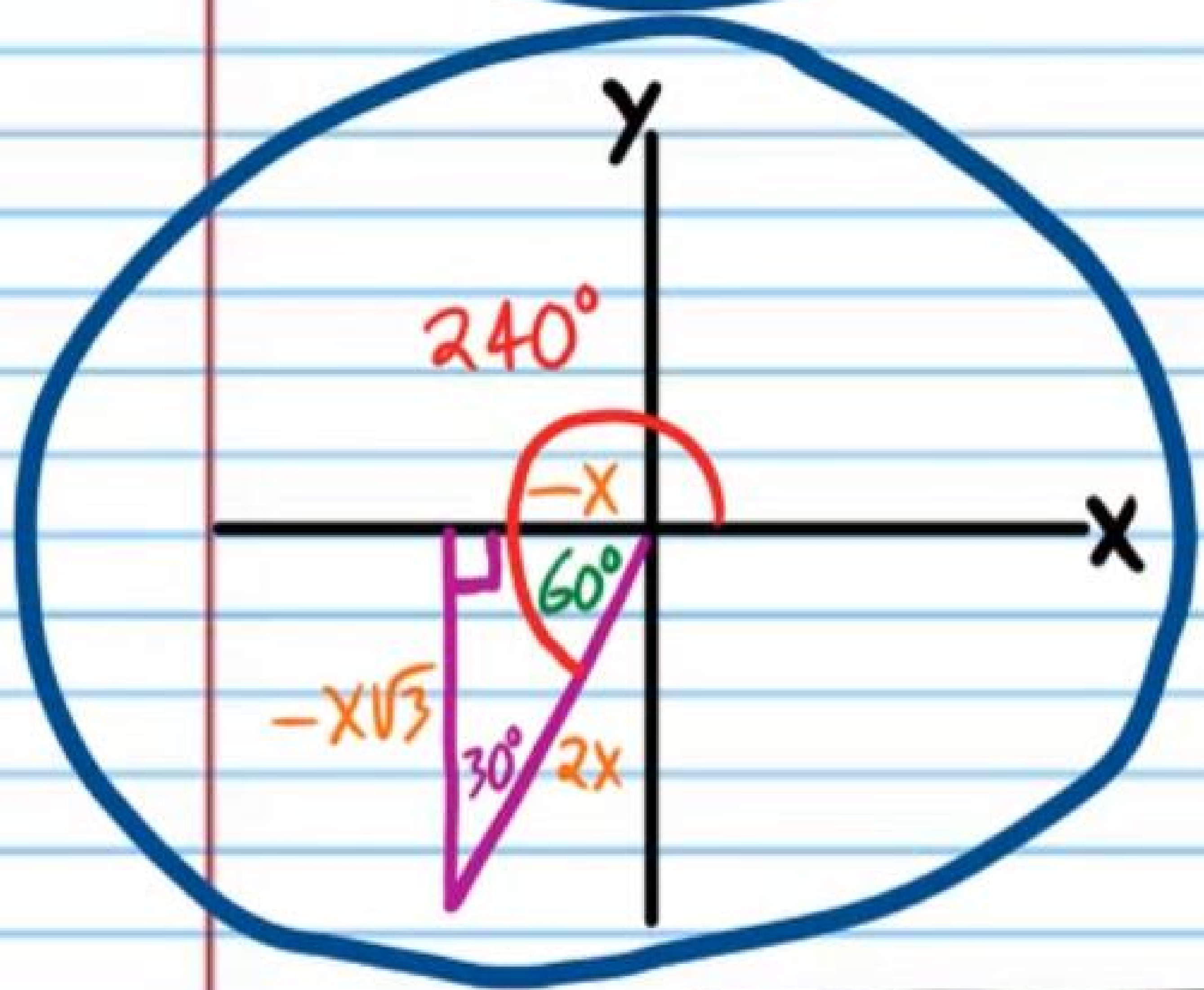
Finding All Solutions of a Trigonometric Equation

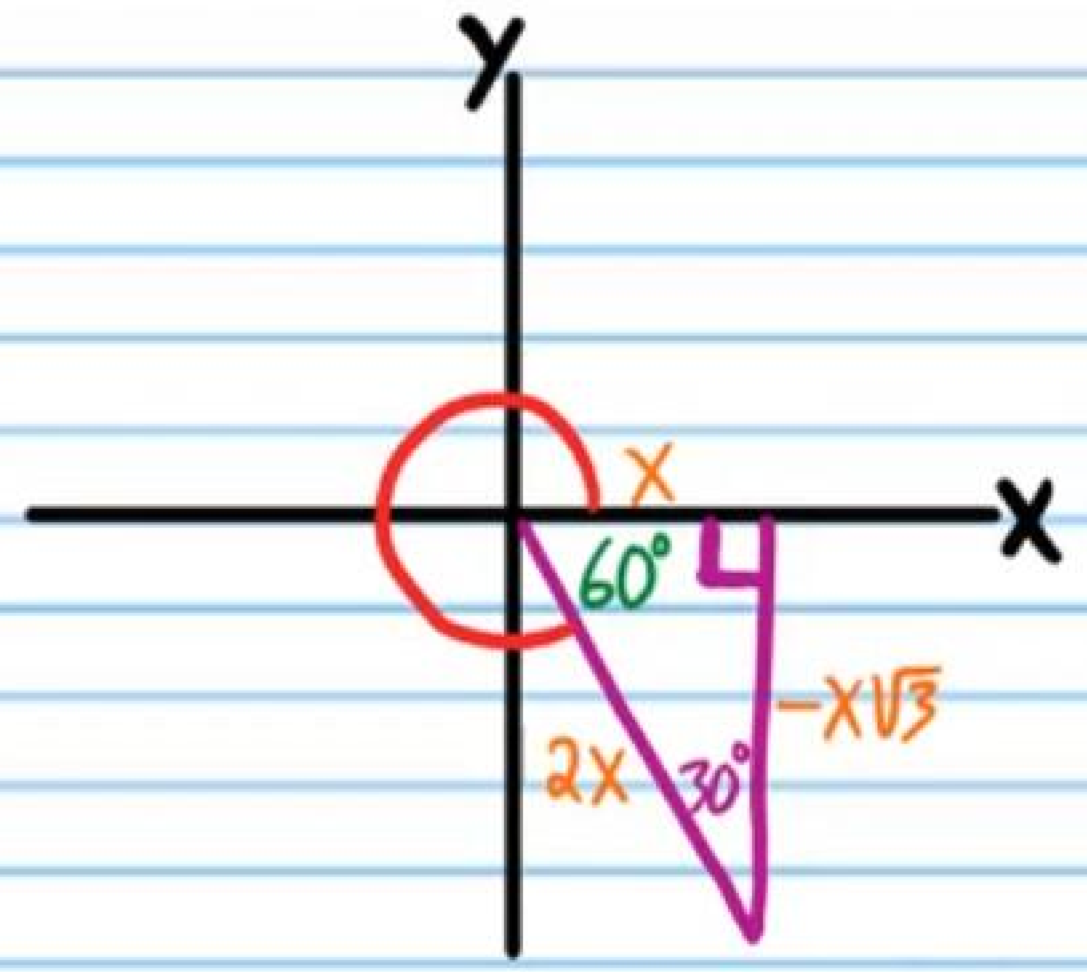
① If $\cos\theta = -\frac{1}{2}$, Find all values of θ.

$$\cos\theta = -\frac{1}{2} = \frac{-x}{2x}$$







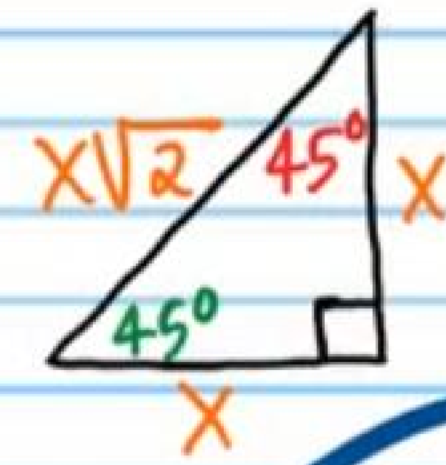


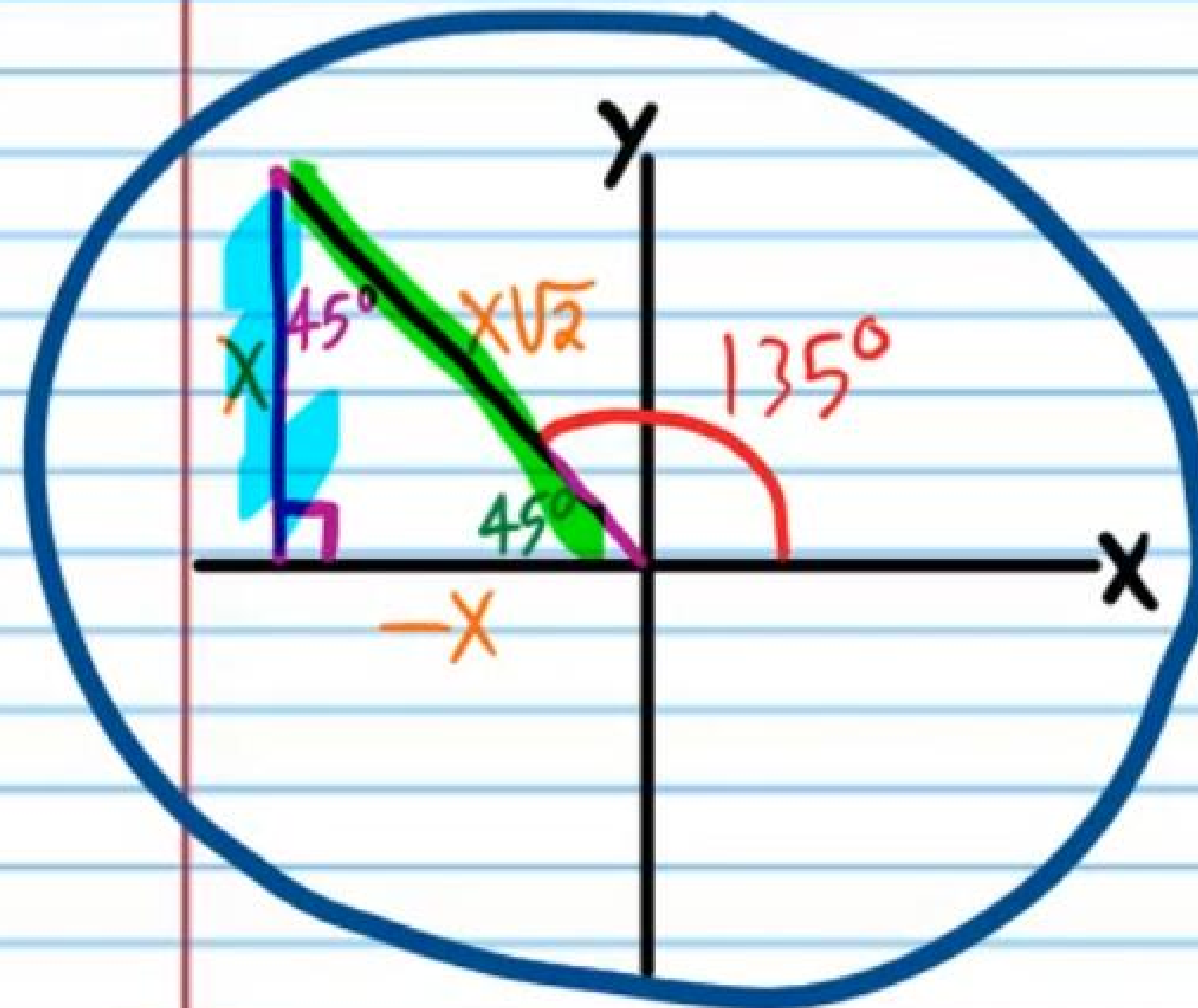
$\theta = 120 + 360n$, n is any integer

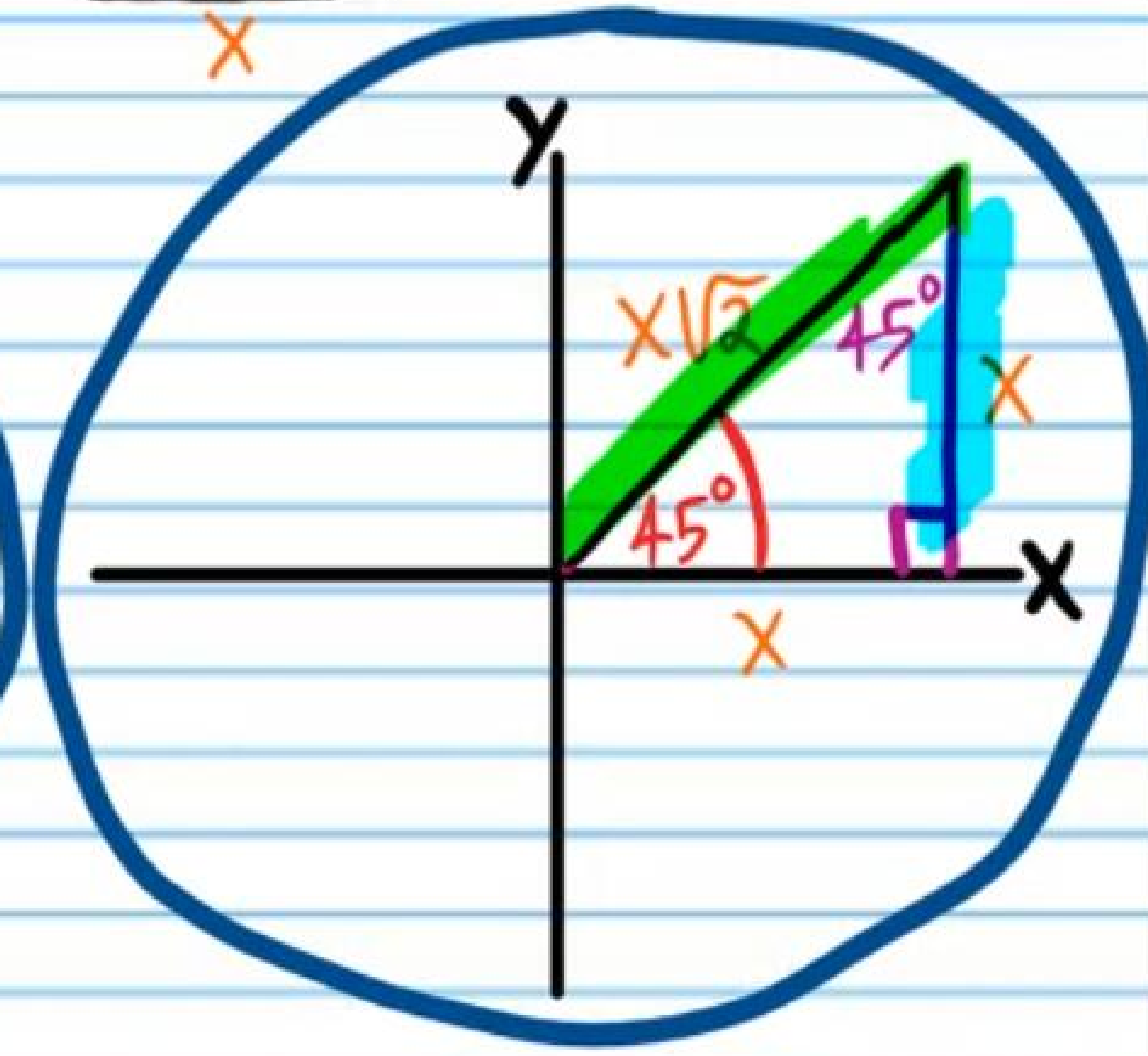
$\theta = 240 + 360n$, n is any integer

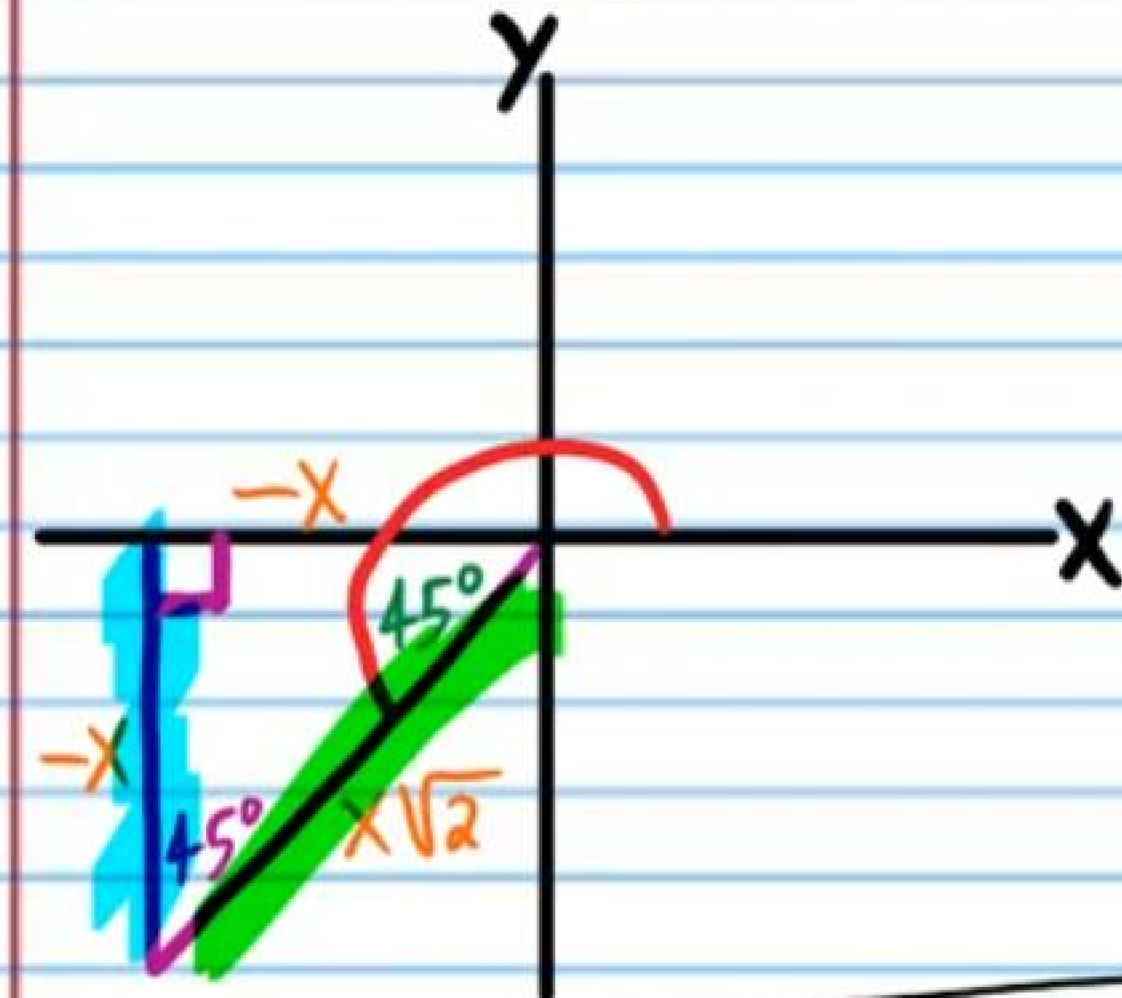
② If $\csc\theta = \sqrt{2}$, Find all values of θ.

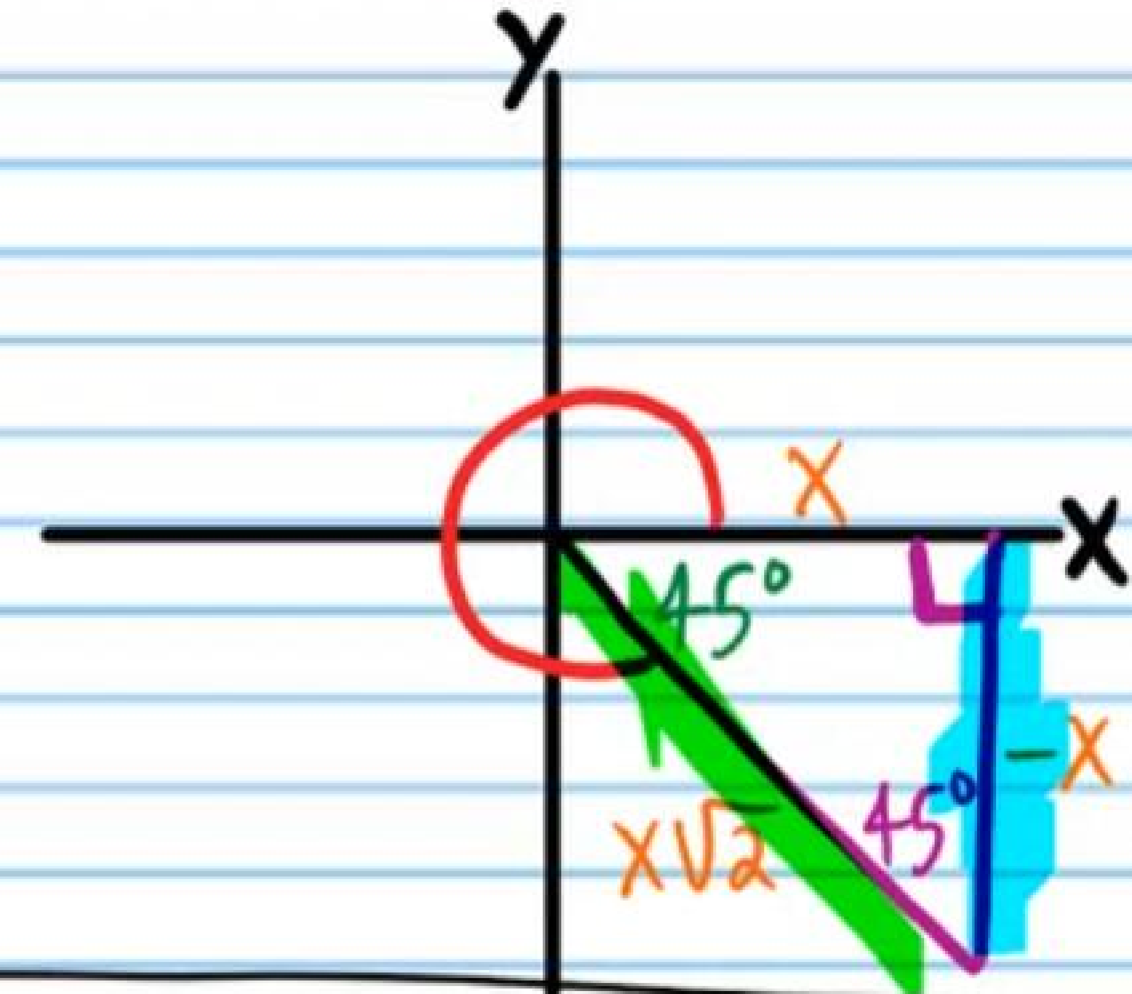
$$\csc\theta = \frac{\sqrt{2}}{1} = \frac{x\sqrt{2}}{x}$$









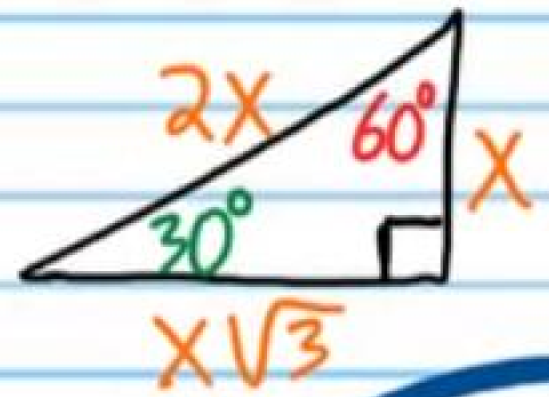


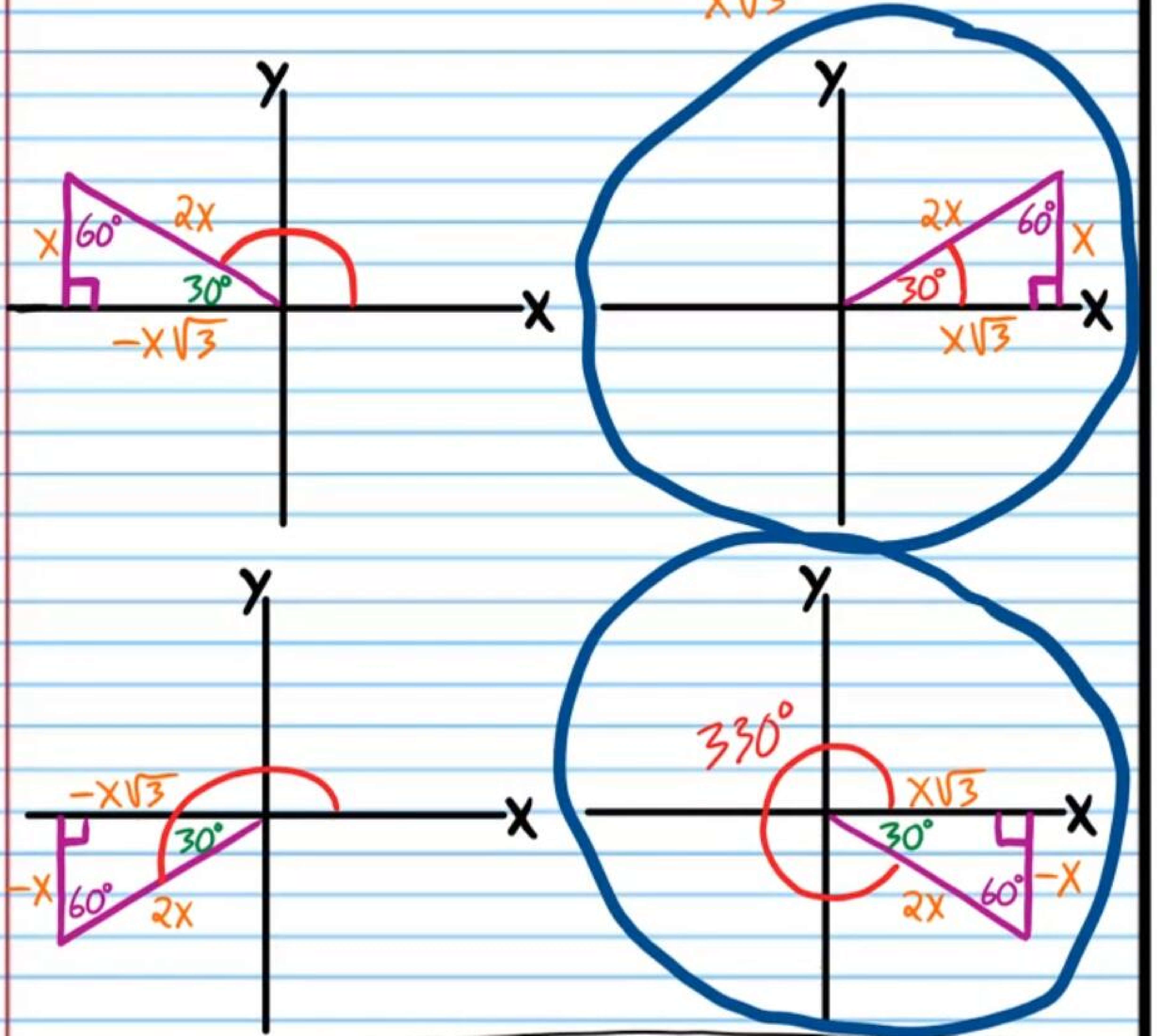
$\theta = 45 + 360n$, n is any integer

$\theta = 135 + 360n$, n is any integer

③ If $\sec\theta = \frac{2}{\sqrt{3}}$, Find all values of θ.

$$\sec\theta = \frac{2}{\sqrt{3}} = \frac{2x}{x\sqrt{3}}$$



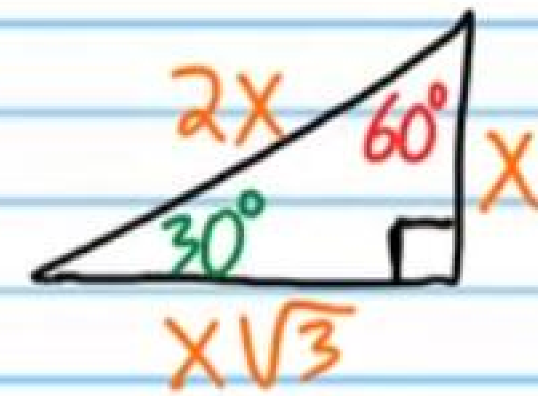


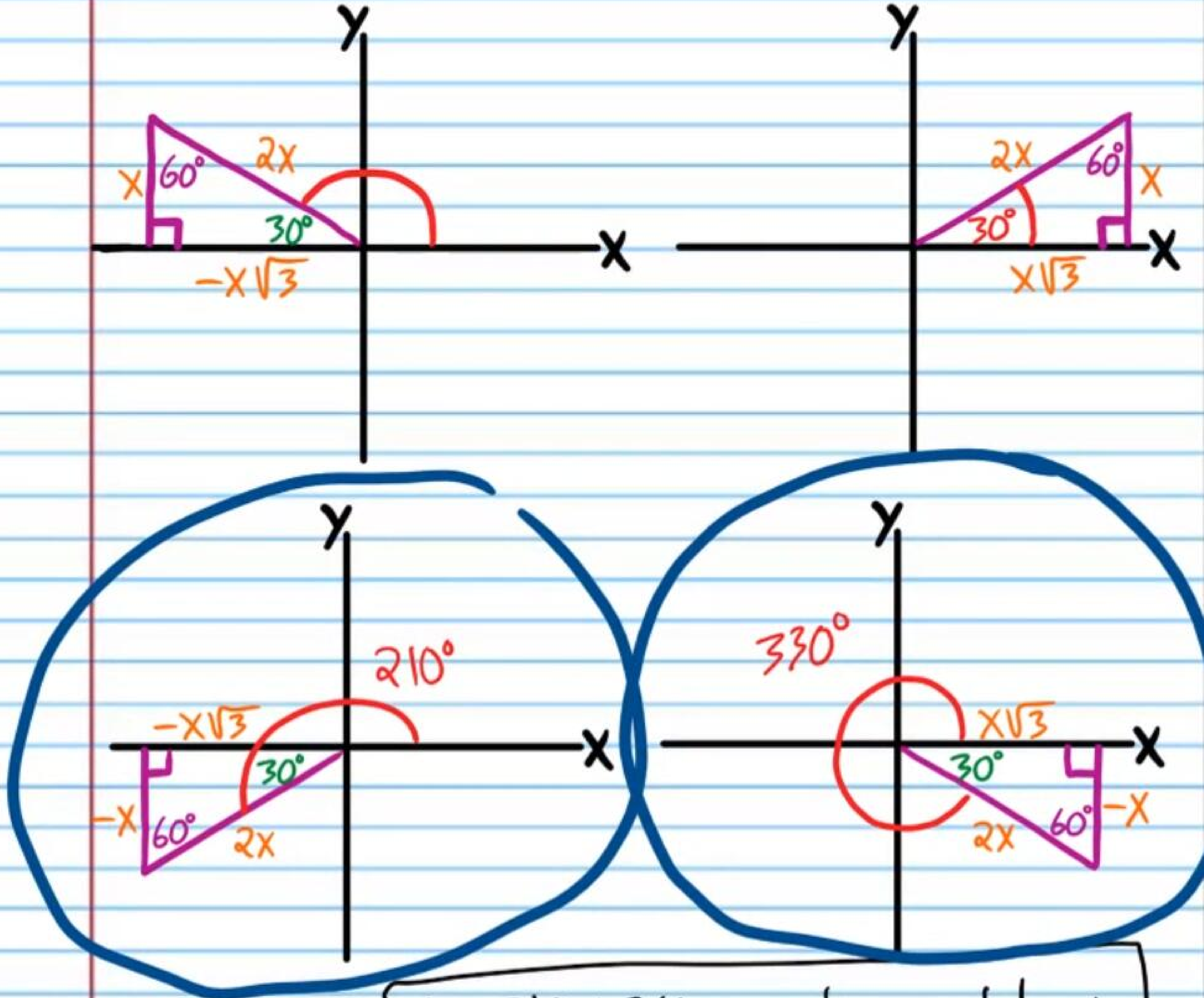
$\theta = 30^\circ + 360n$, n is any integer

$\theta = 330^\circ + 360n$, n is any integer

④ If $\sin\theta = -\frac{1}{2}$, Find all values of θ.

$$\sin\theta = -\frac{1}{2} = \frac{-x}{2x}$$



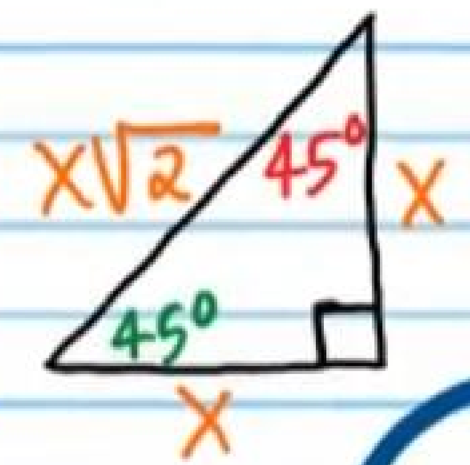


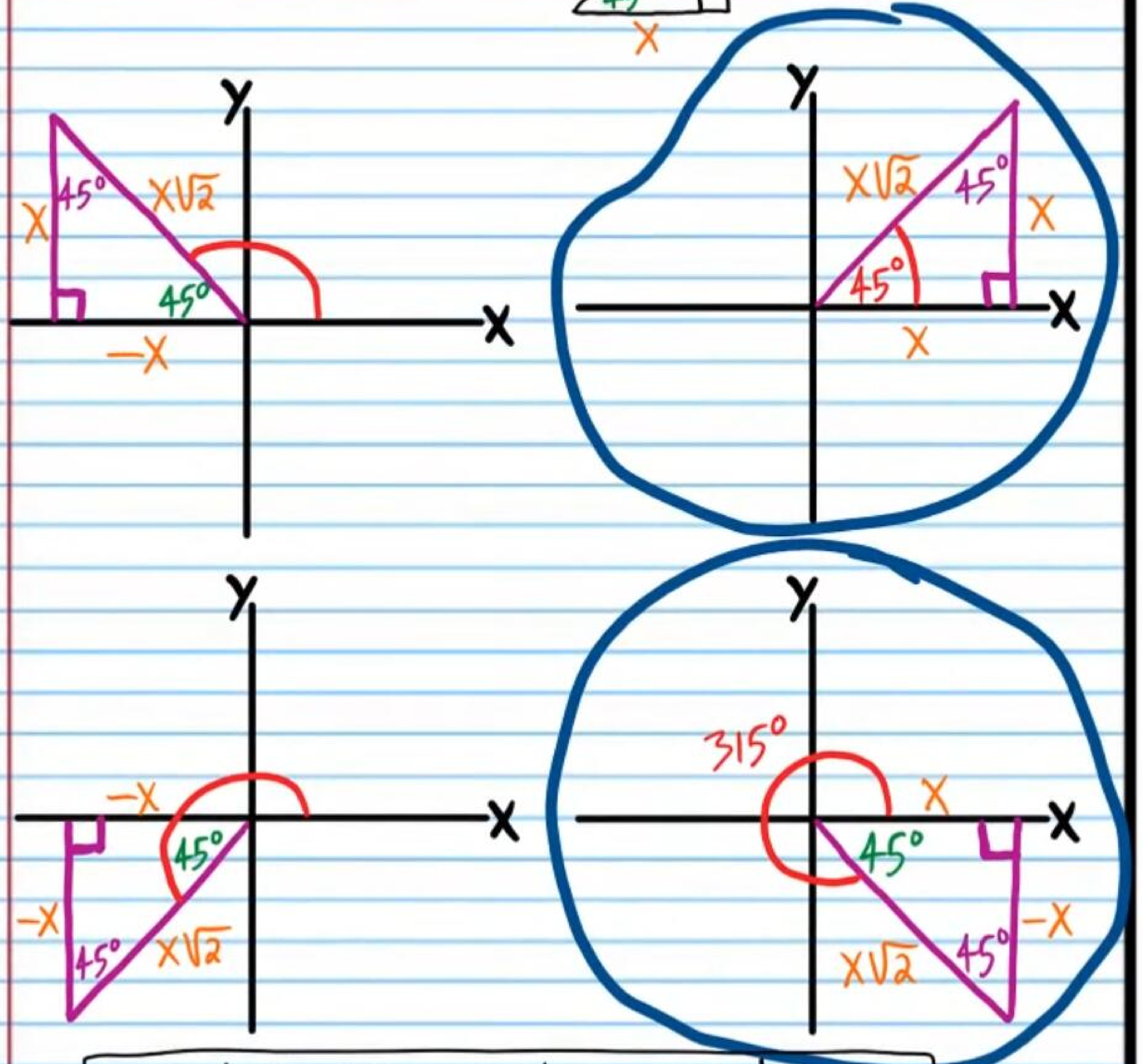
$\theta = 210 + 360n$, n is any integer

$\theta = 330 + 360n$, n is any integer

⑤ If $\cos\theta = \frac{1}{\sqrt{2}}$, Find all values of θ.

$$\cos\theta = \frac{1}{\sqrt{2}} = \frac{x}{x\sqrt{2}}$$



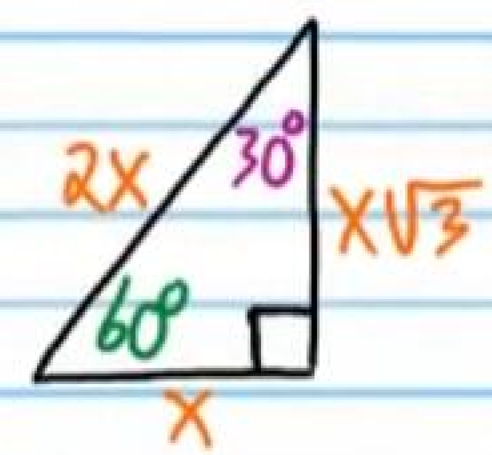


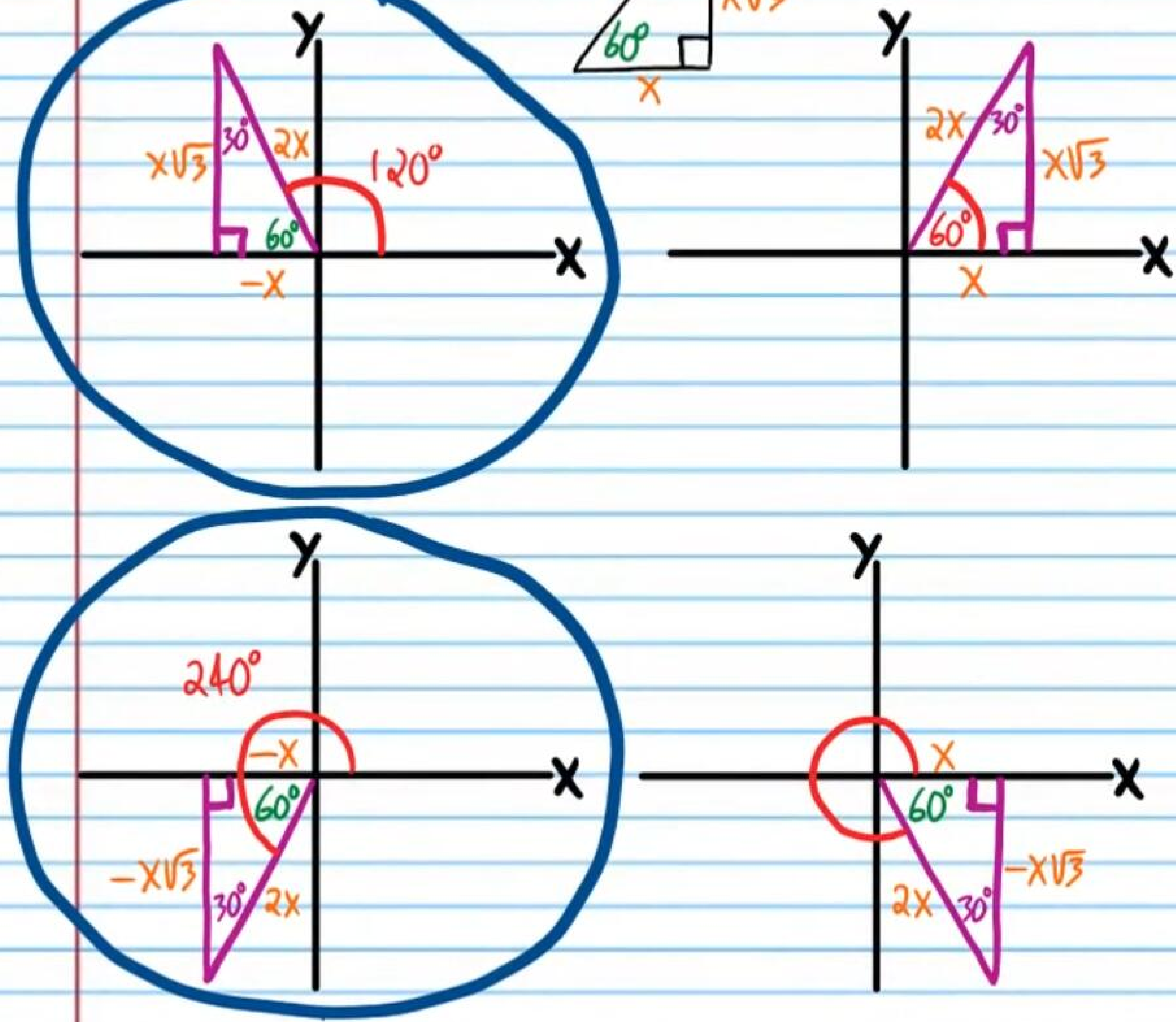
$\theta = 45 + 360n$, n is any integer

$\theta = 315 + 360n$, n is any integer

⑥ If $\sec\theta = -2$, Find all values of θ.

$$\sec\theta = \frac{2}{-1} = \frac{2x}{-x}$$



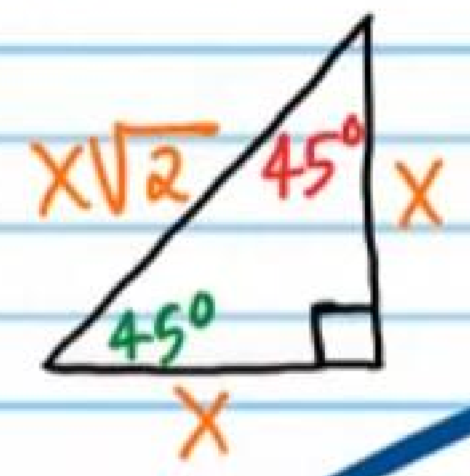


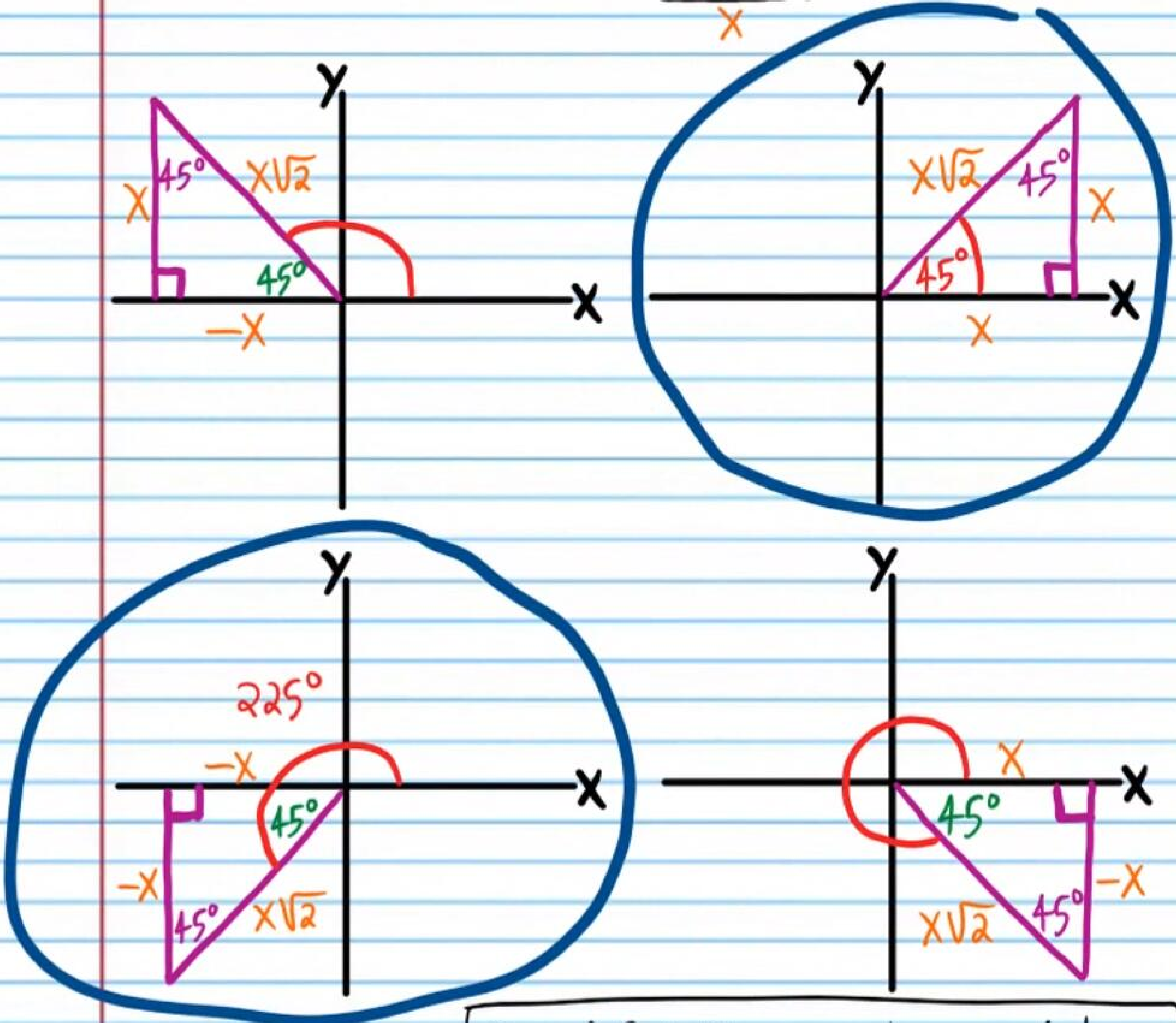
$\theta = 120 + 360n$, n is any integer

$\theta = 240 + 360n$, n is any integer

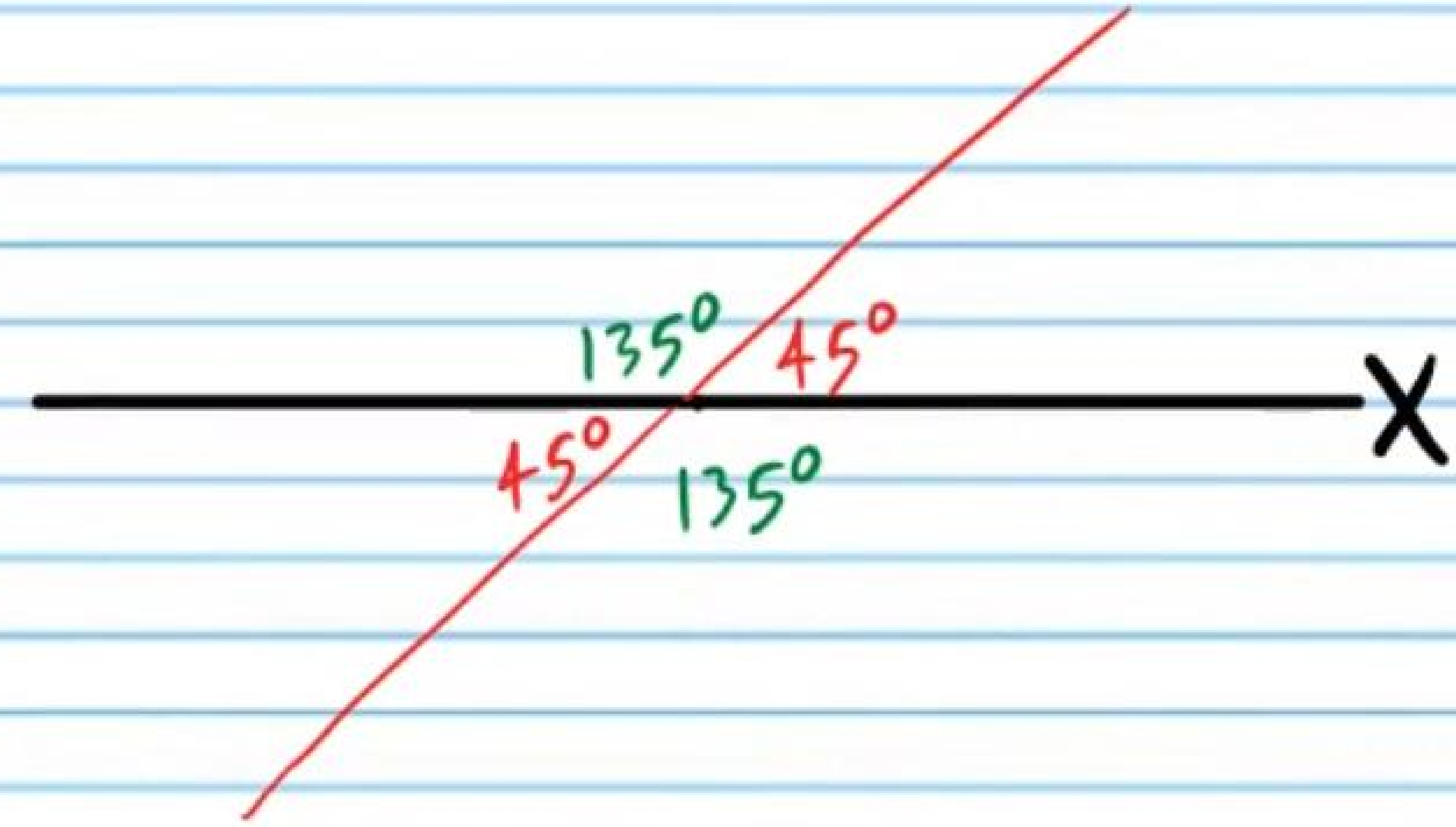
• ⑦ If $\tan\theta = 1$, find all values of θ.

$$\tan\theta = \frac{1}{1} = \frac{x}{x}$$



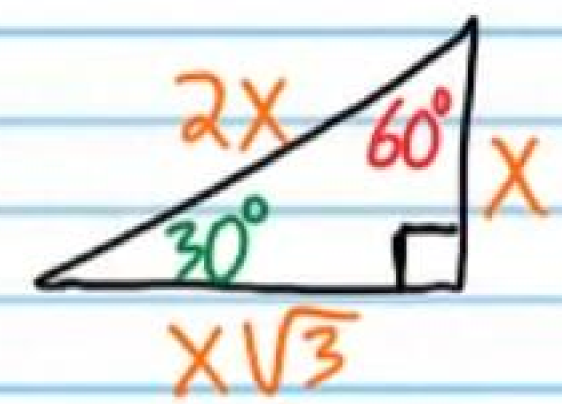


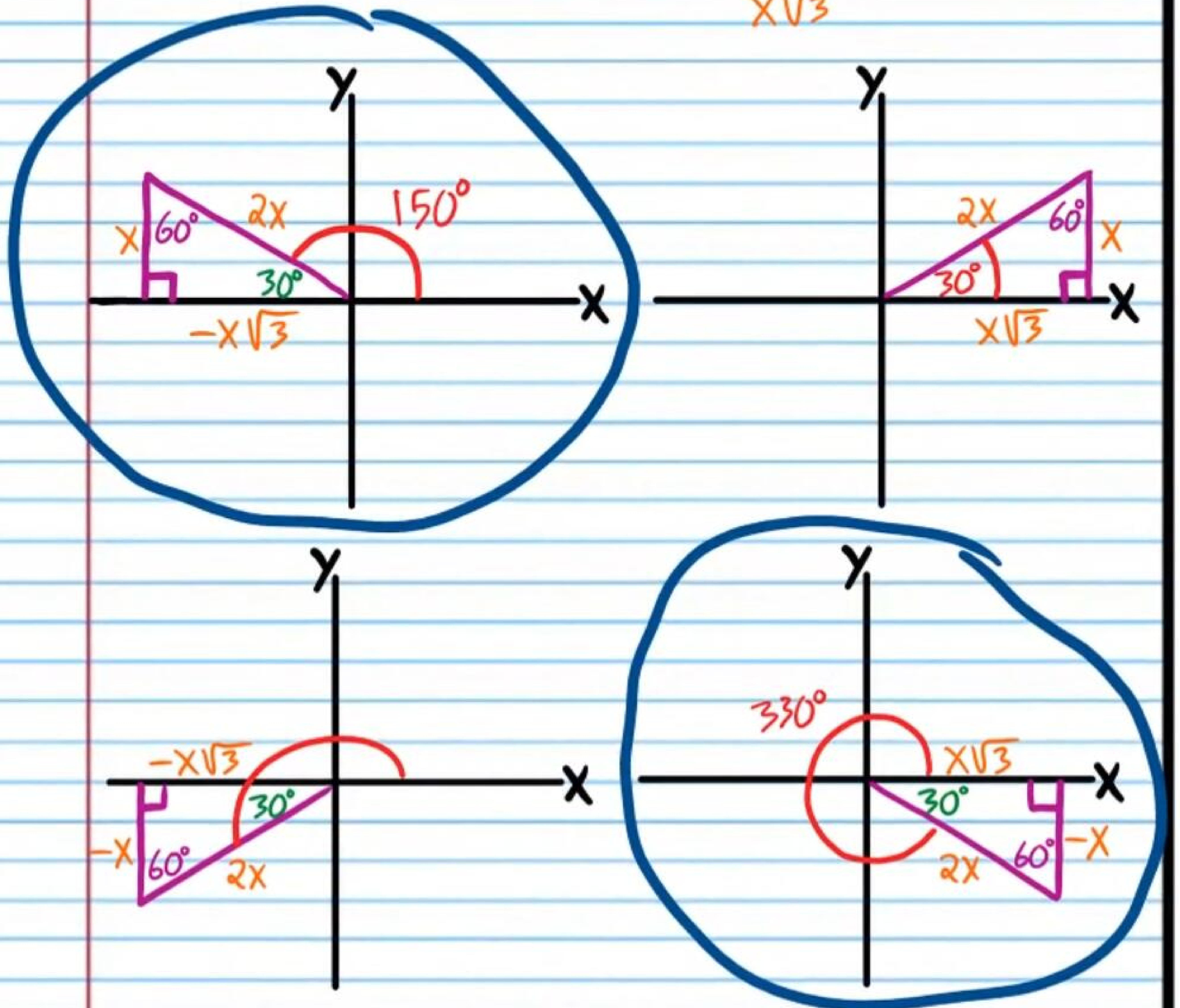
$\boxed{\theta = 45 + 180n, \text{ n is any integer}}$



⑧ If $\cot\theta = -\sqrt{3}$, find all values of θ.

$$\cot\theta = -\frac{\sqrt{3}}{1} = -\frac{x\sqrt{3}}{x}$$

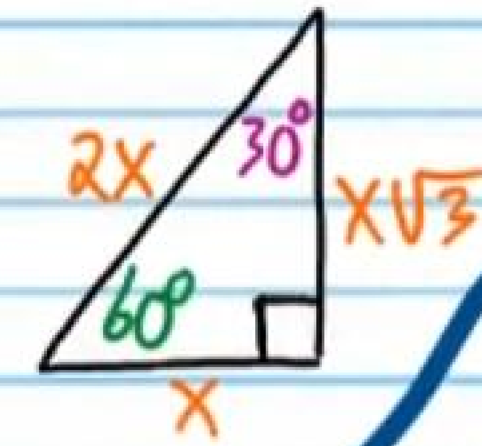


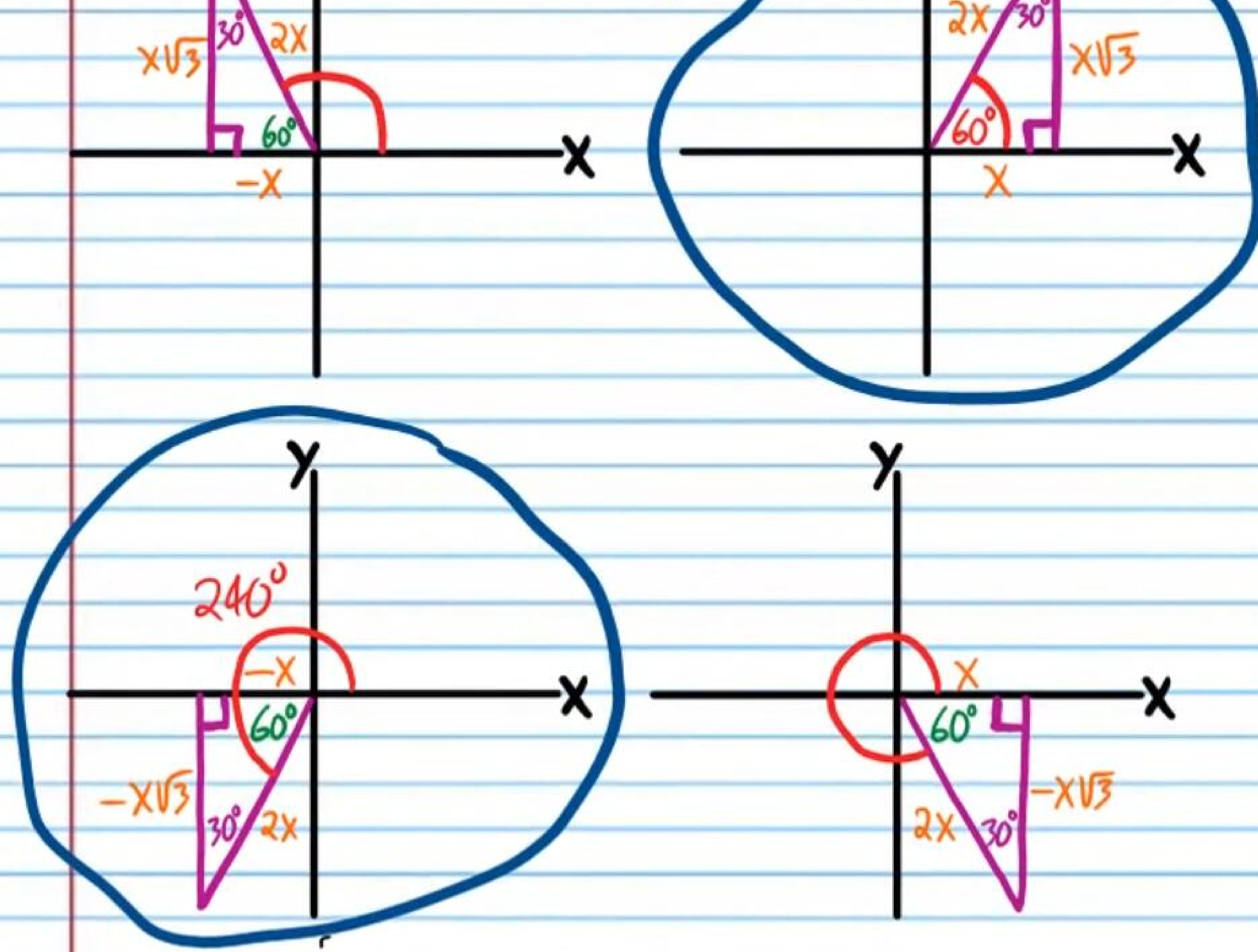


$$\boxed{\theta = 150 + 180n, \text{ n is any integer}}$$

⑨ If $\tan\theta = \sqrt{3}$, find all values of θ.

$$\tan\theta = \frac{\sqrt{3}}{1} = \frac{x\sqrt{3}}{x}$$

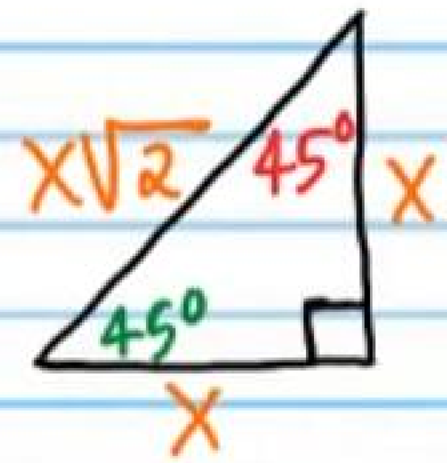


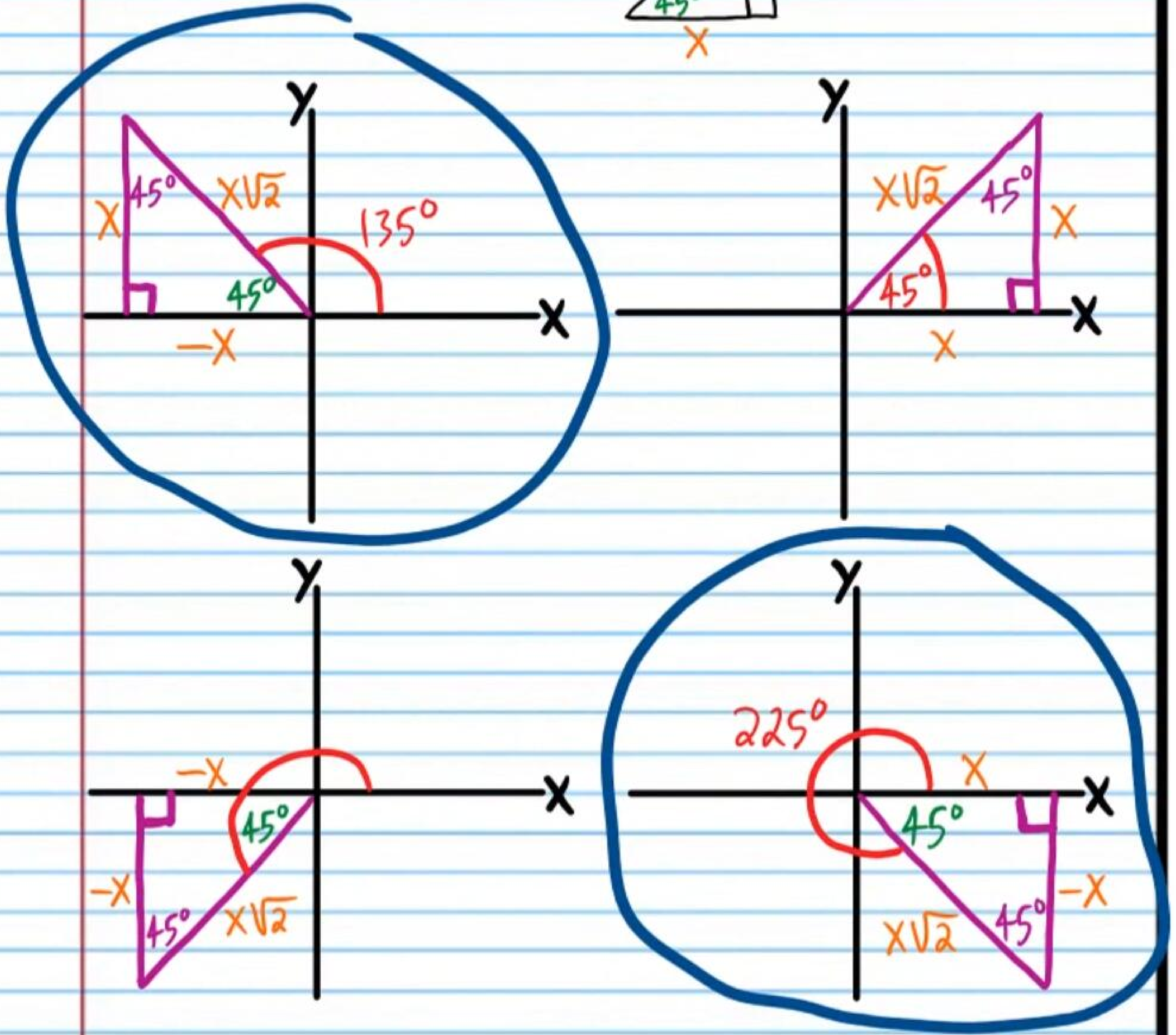


$$\boxed{\theta = 60 + 180n, \text{ n is any integer}}$$

⑩ If $\cot\theta = -1$, find all values of θ.

$\cot\theta = -\frac{1}{1} = -\frac{x}{x}$





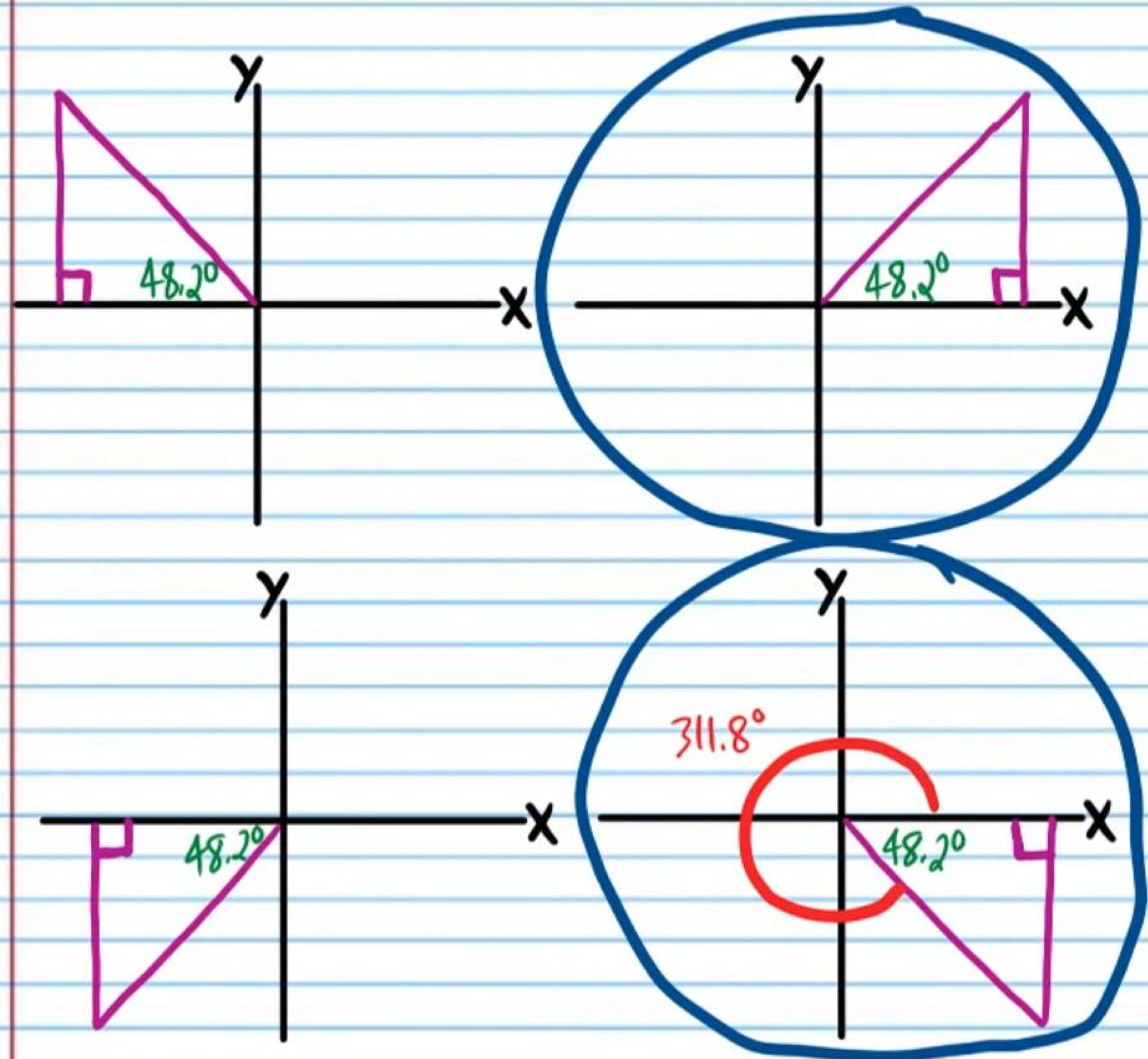
$\boxed{\theta = 135 + 180n,\ n \text{ is any integer}}$

Finding Angle Measures with A Calculator

★ Make sure that your calculator is in degree mode.

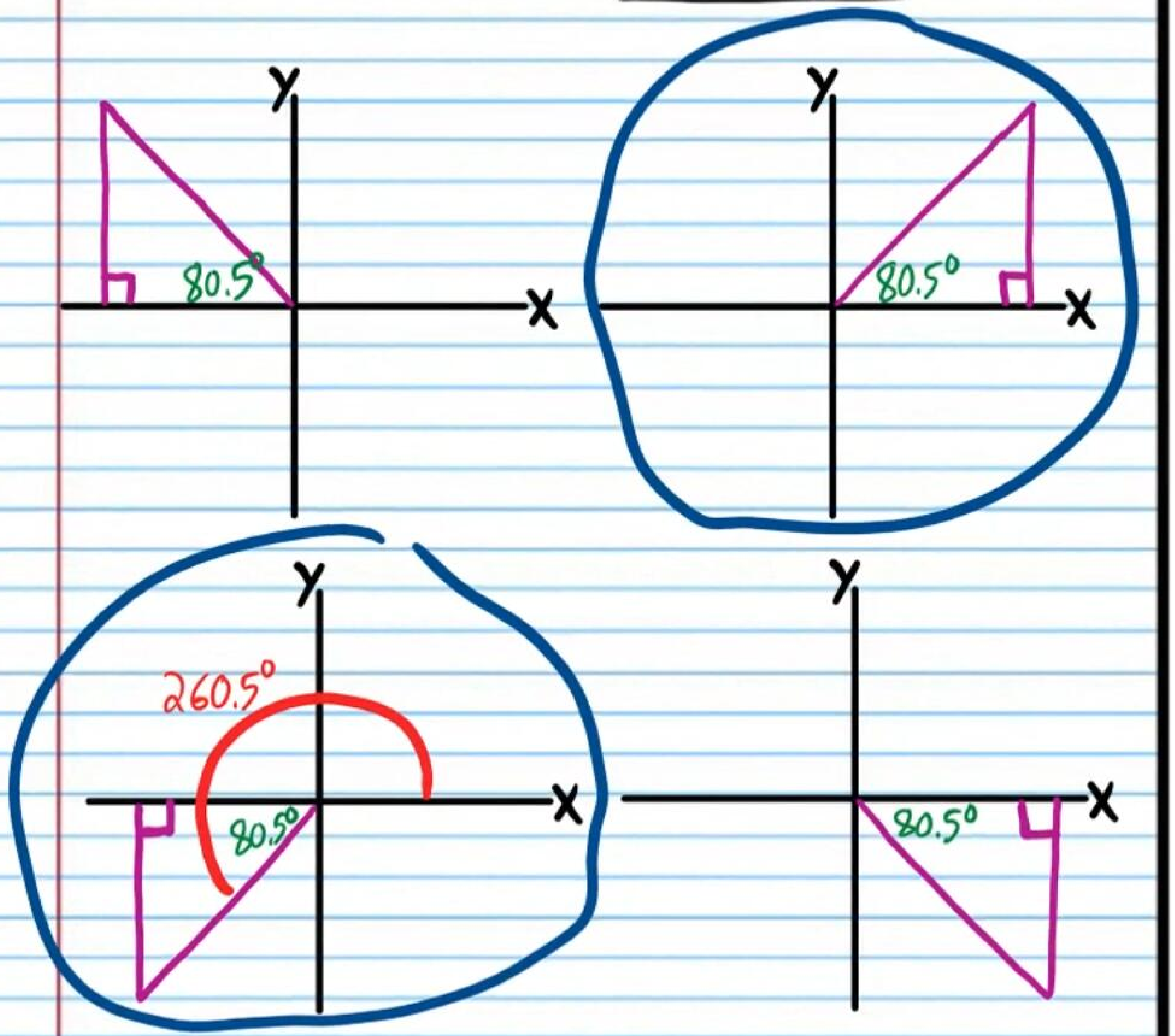
• ⑪ If $\cos\theta = \frac{2}{3}$, and $0° < \theta \leq 360°$, find the values of θ.

$\theta = \cos^{-1}\left(\frac{2}{3}\right) \approx \boxed{48.2°,\ 311.8°}$



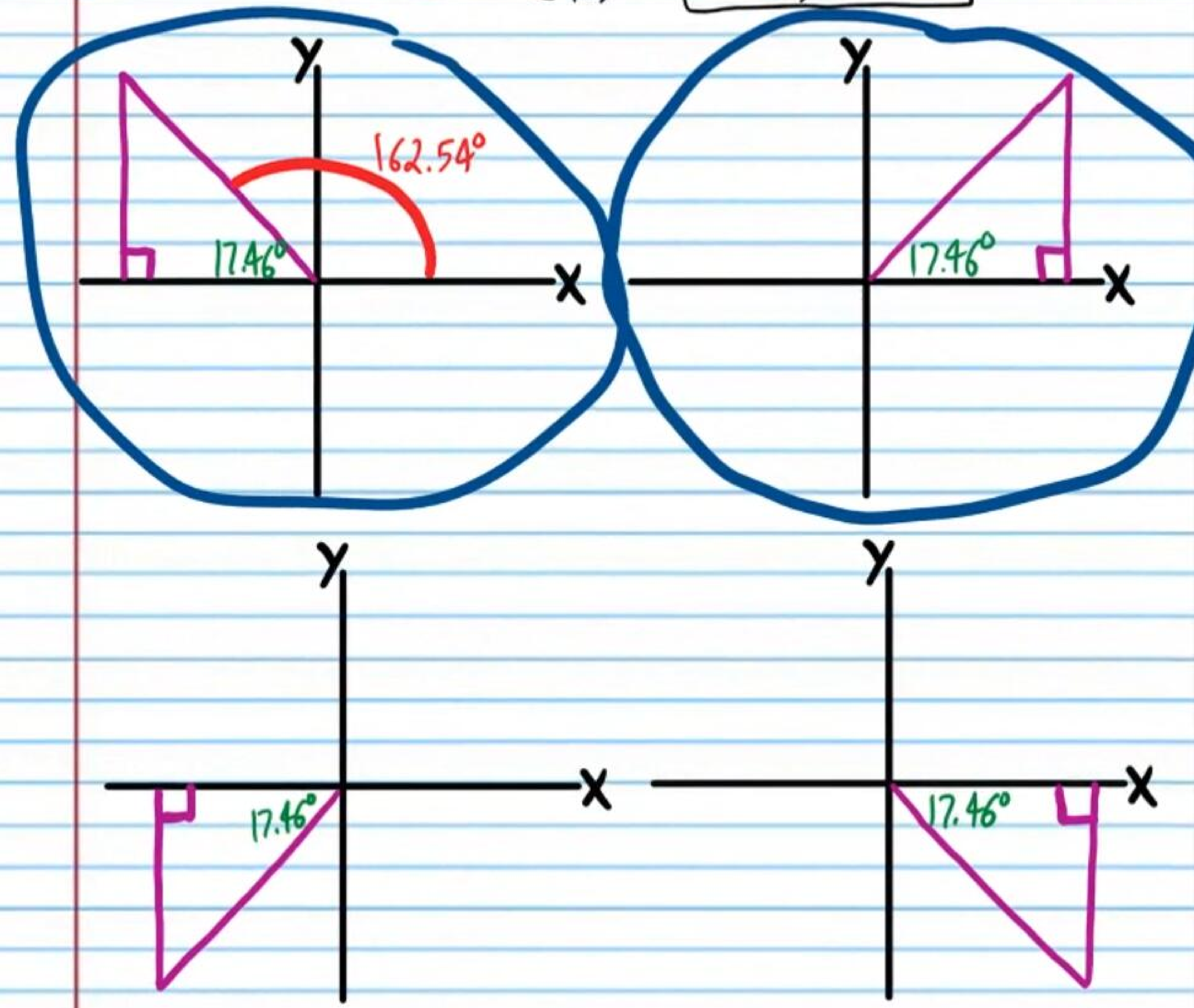
(12) If $\tan\theta = 6$, and $0° < \theta \le 360°$, Find the values of θ.

$$\theta = \tan^{-1} 6 \approx \boxed{80.5°, 260.5°}$$



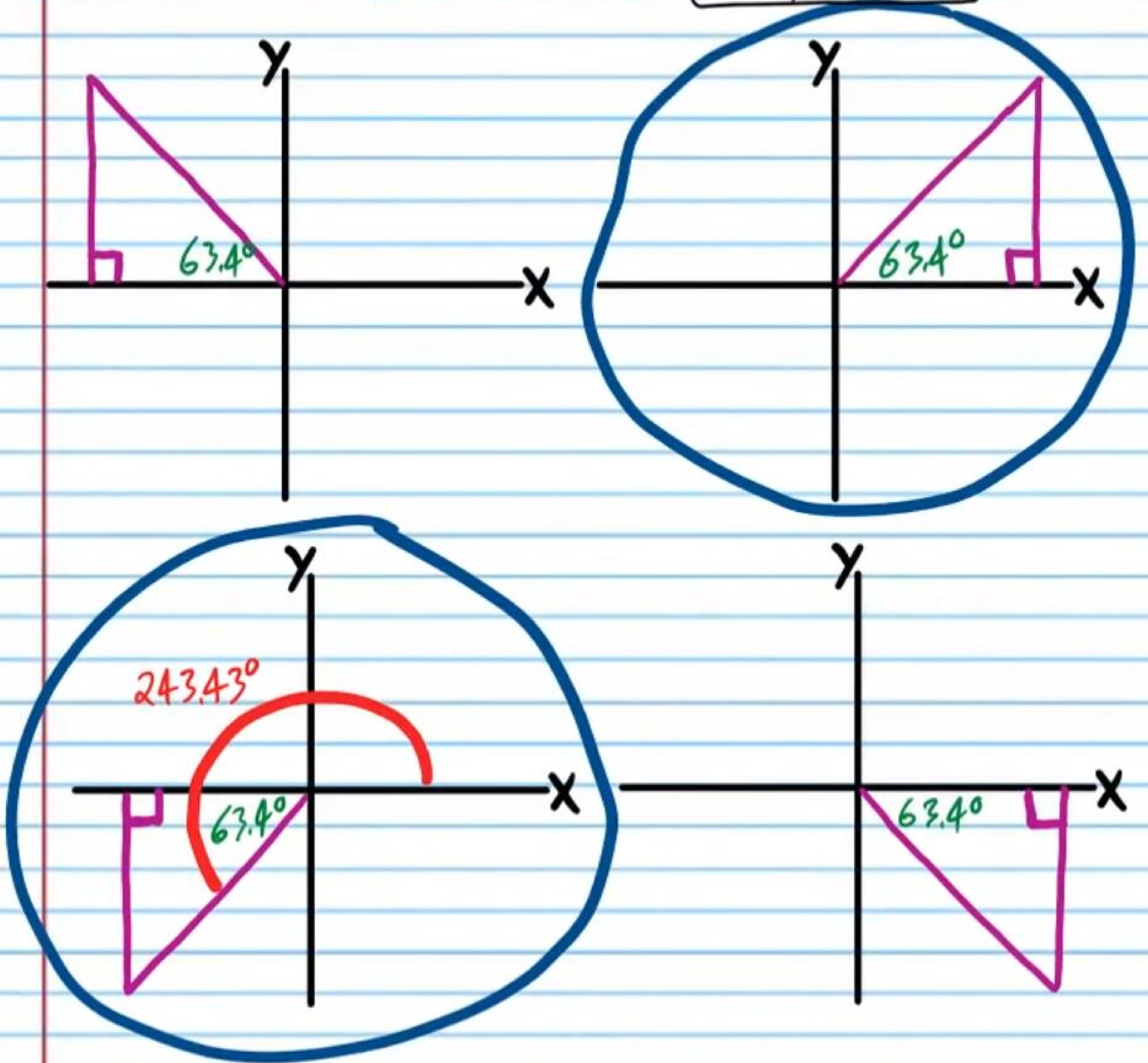
(13) If $\sin\theta = \frac{3}{10}$, and $0° < \theta \le 360°$, Find the values of θ.

$$\theta = \sin^{-1}\left(\frac{3}{10}\right) \approx \boxed{17.46°, 162.54°}$$



(14) If $\tan\theta = 2$, and $0° < \theta \leq 360°$, Find the values of θ.

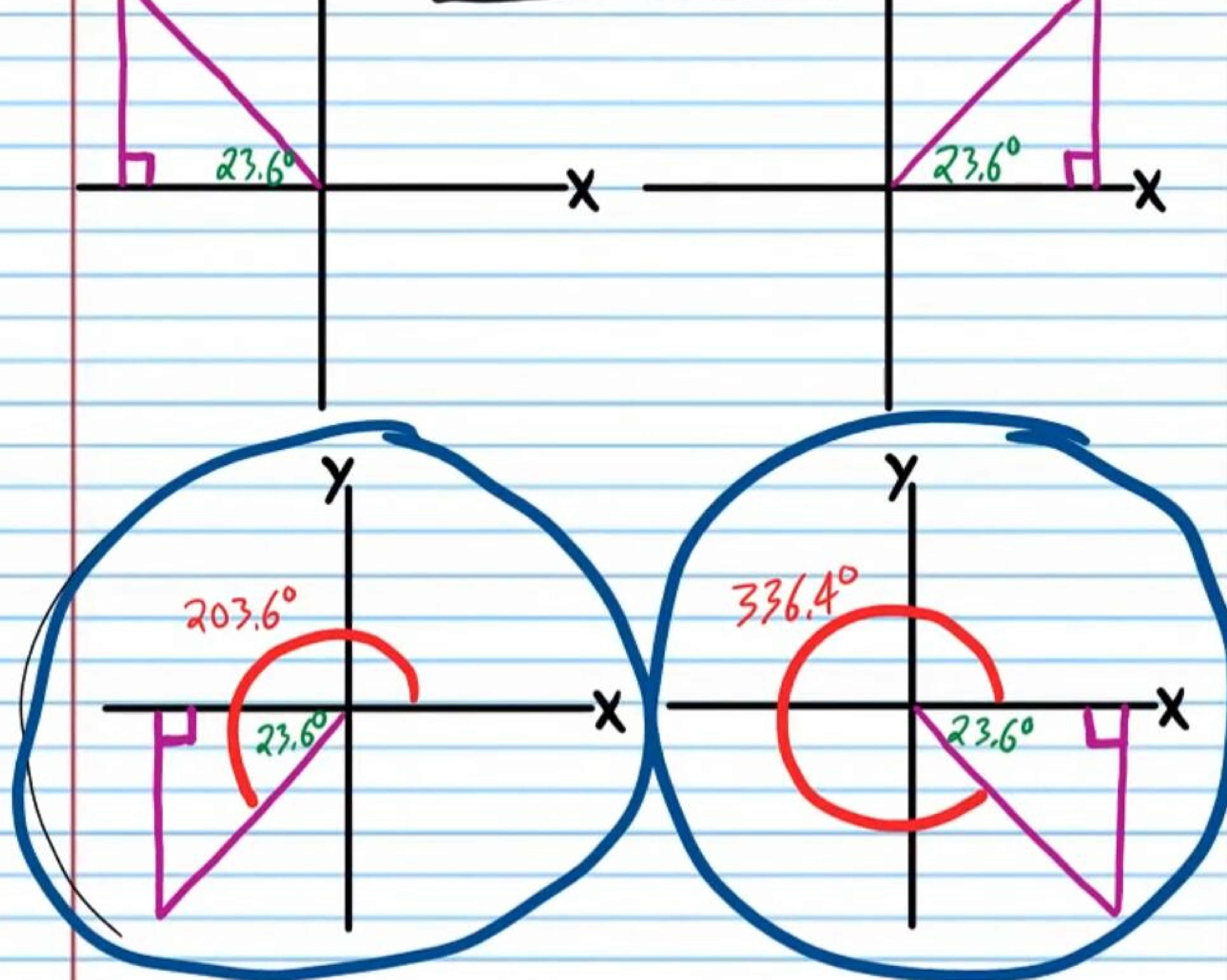
$$\theta = \tan^{-1} 2 \approx \boxed{63.4°,\ 243.43°}$$



(15) If $\sin\theta = -0.4$, and $0° < \theta \leq 360°$, Find the values of θ.

$$\theta = \sin^{-1}(-0.4) \not\approx -23.6°$$

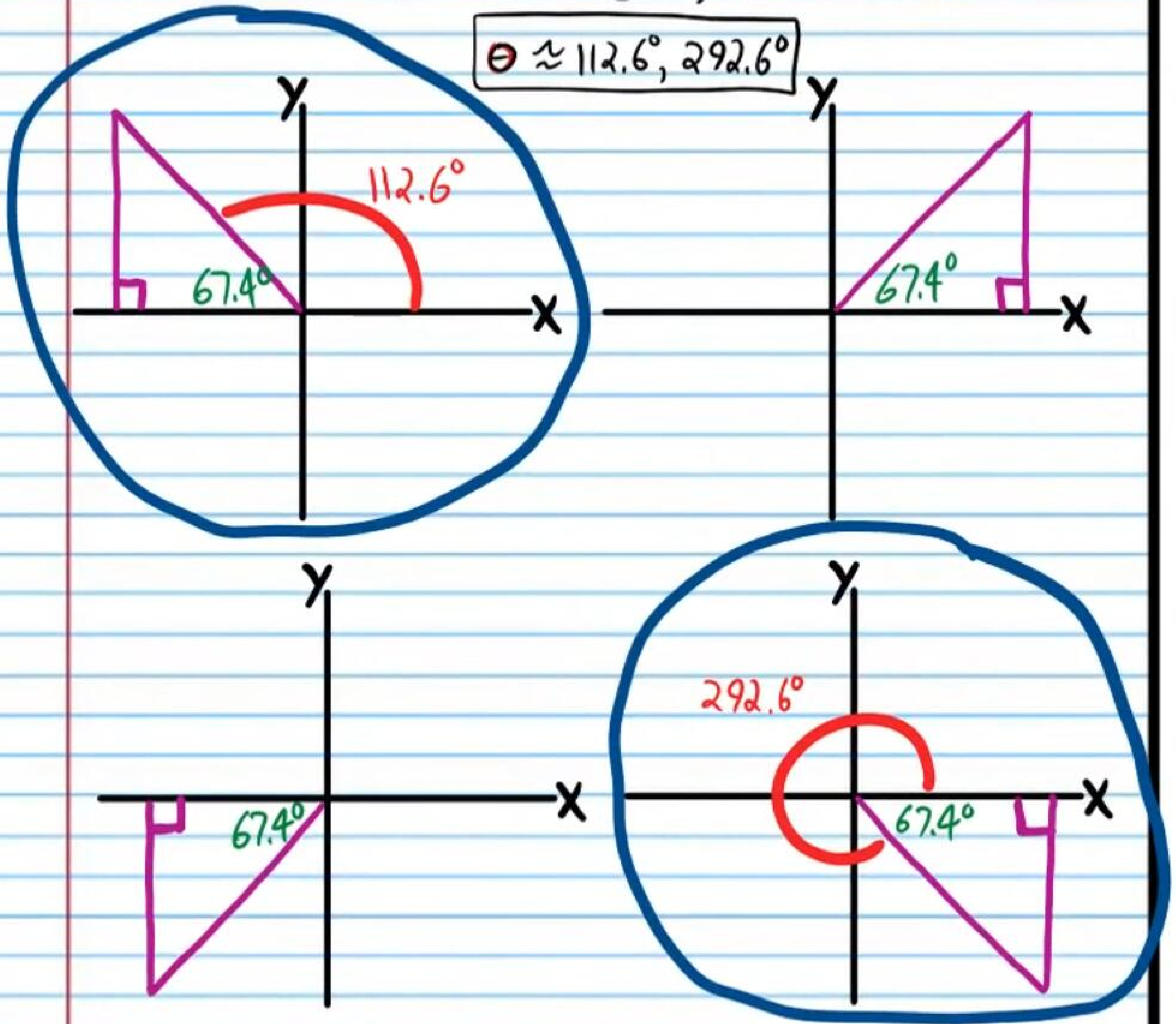
$$\boxed{\theta \approx 203.6°,\ 336.4°}$$



16. If $\tan\theta = -2.4$, and $0° < \theta \leq 360°$, Find the values of θ.

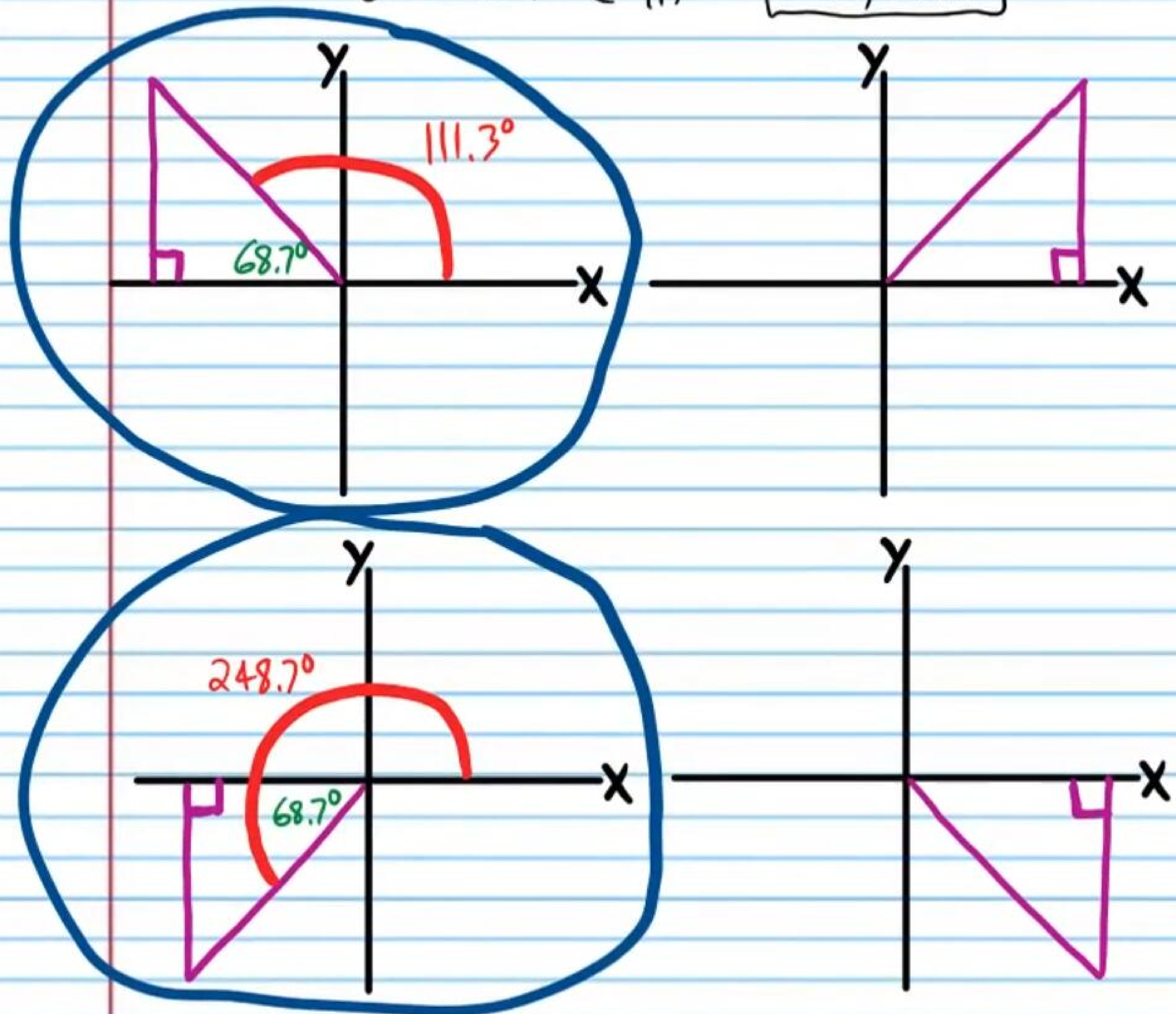
$$\theta = \tan^{-1}(-2.4) \not\approx -67.4°$$

$$\boxed{\theta \approx 112.6°,\ 292.6°}$$



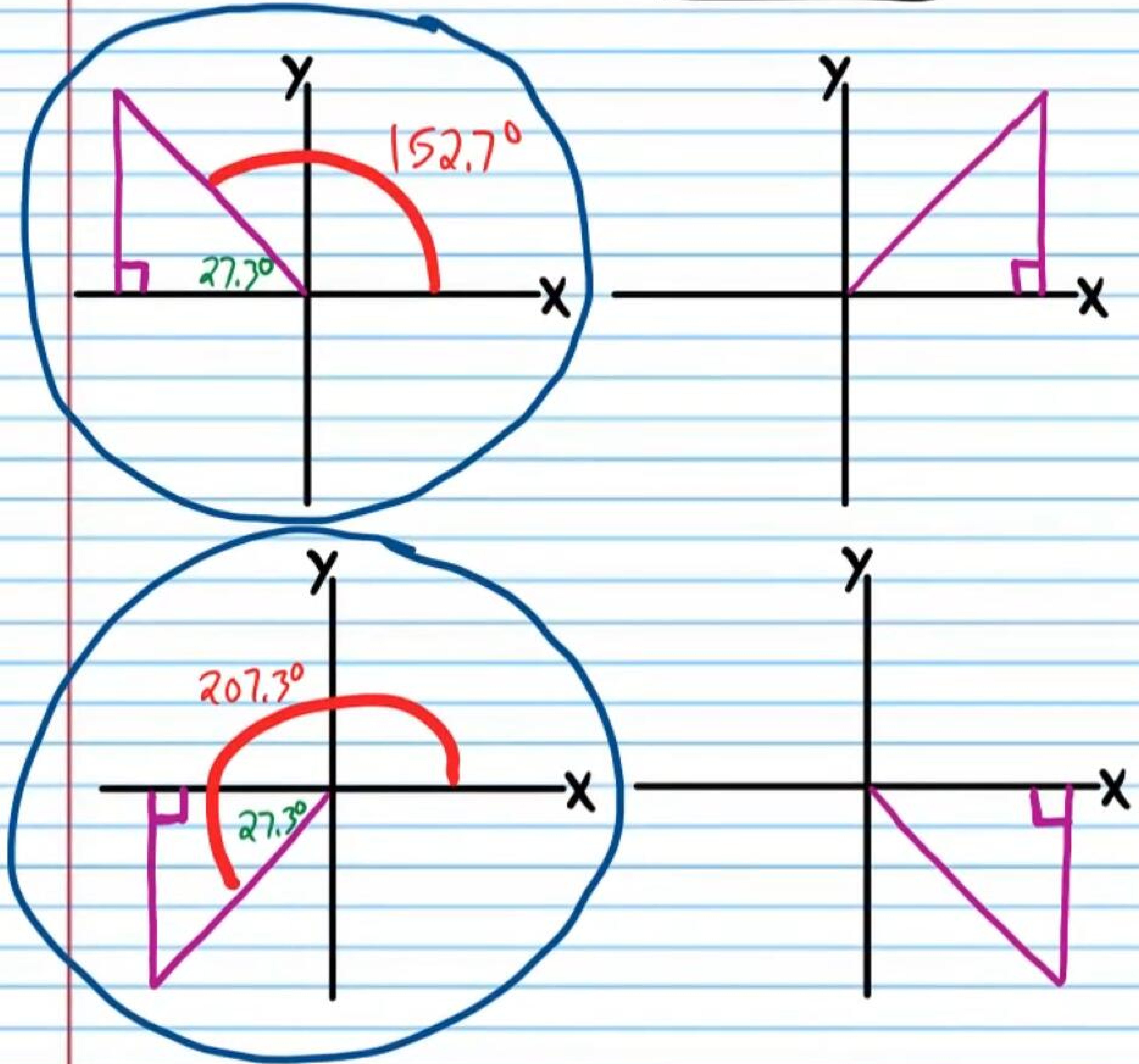
17. If $\cos\theta = -\frac{4}{11}$, and $0° < \theta \leq 360°$, Find the values of θ.

$$\theta = \cos^{-1}\left(-\frac{4}{11}\right) \approx \boxed{111.3°,\ 248.7°}$$



18) If $\cos\theta = -\frac{8}{9}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ.

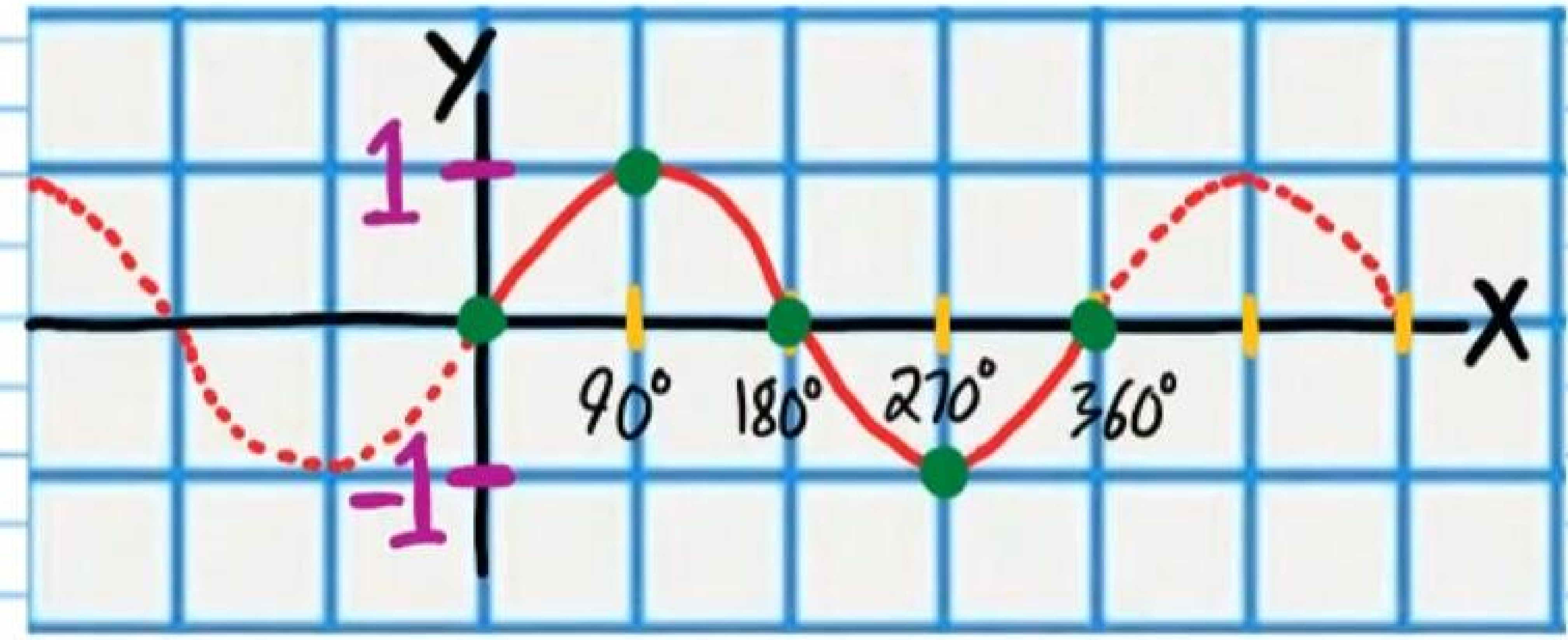
$$\theta = \cos^{-1}\left(-\frac{8}{9}\right) \approx \boxed{152.7^\circ,\ 207.3^\circ}$$



Finding the Measures of Quadrantal Angles That Correspond to a Trigonometric Ratio

19) If $\sin\theta = 1$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ.

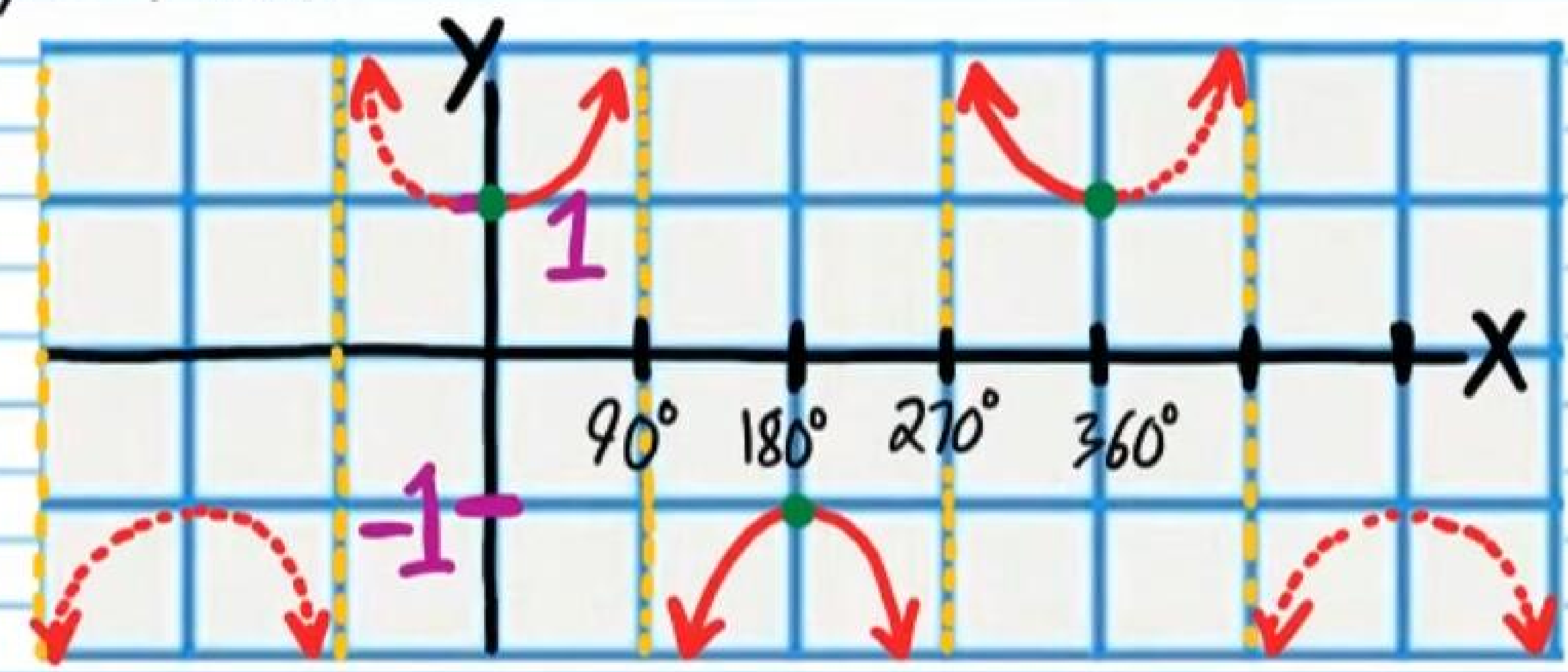
$y = \sin x$



$$\boxed{\theta = 90^\circ}$$

20) If $\sec\theta = 1$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ.

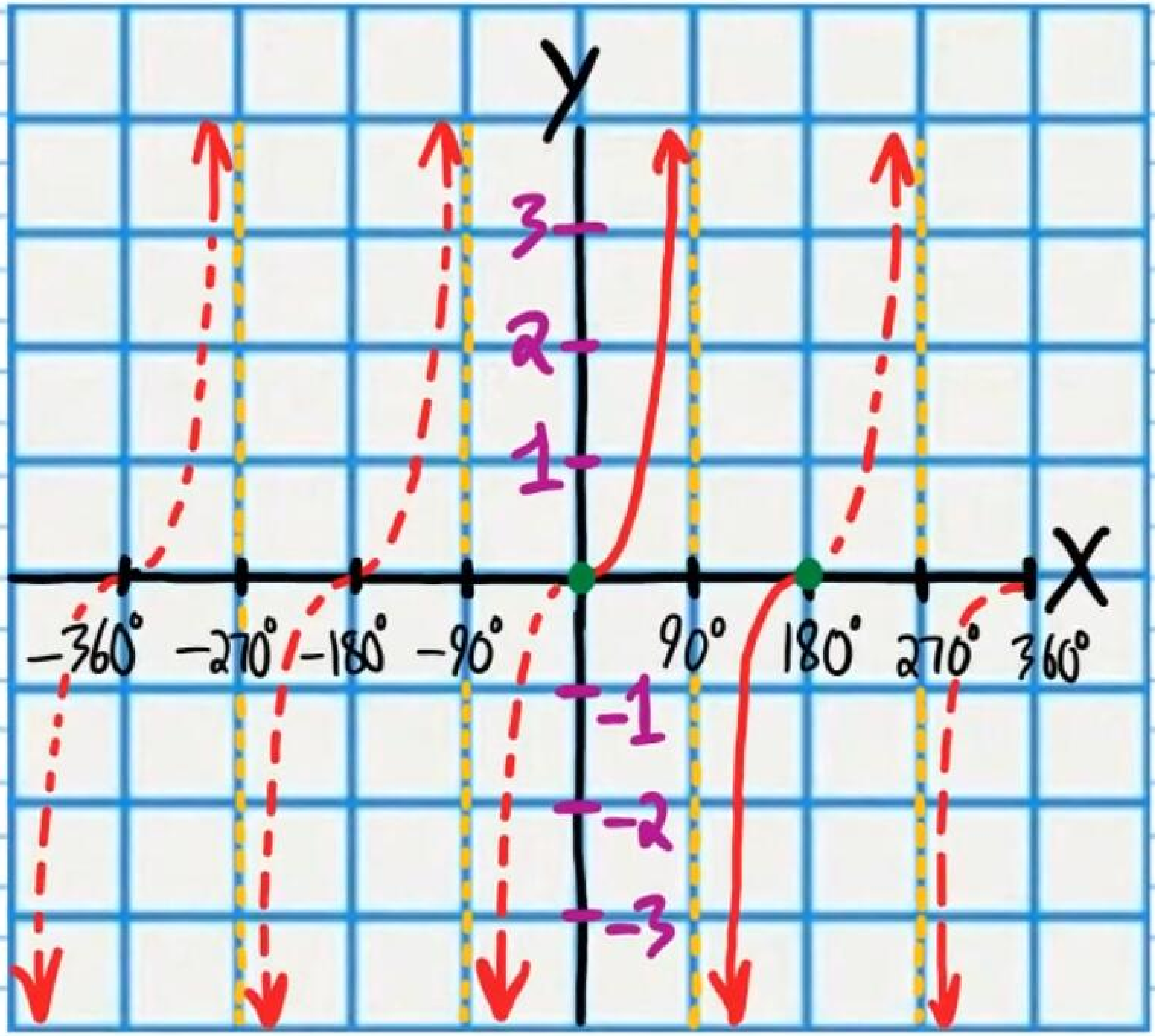
$y = \sec\theta$



$$\boxed{\theta = 360^\circ}$$

(21) If $\tan\theta = 0$, and $0° < \theta \leq 360°$, Find the values of θ.

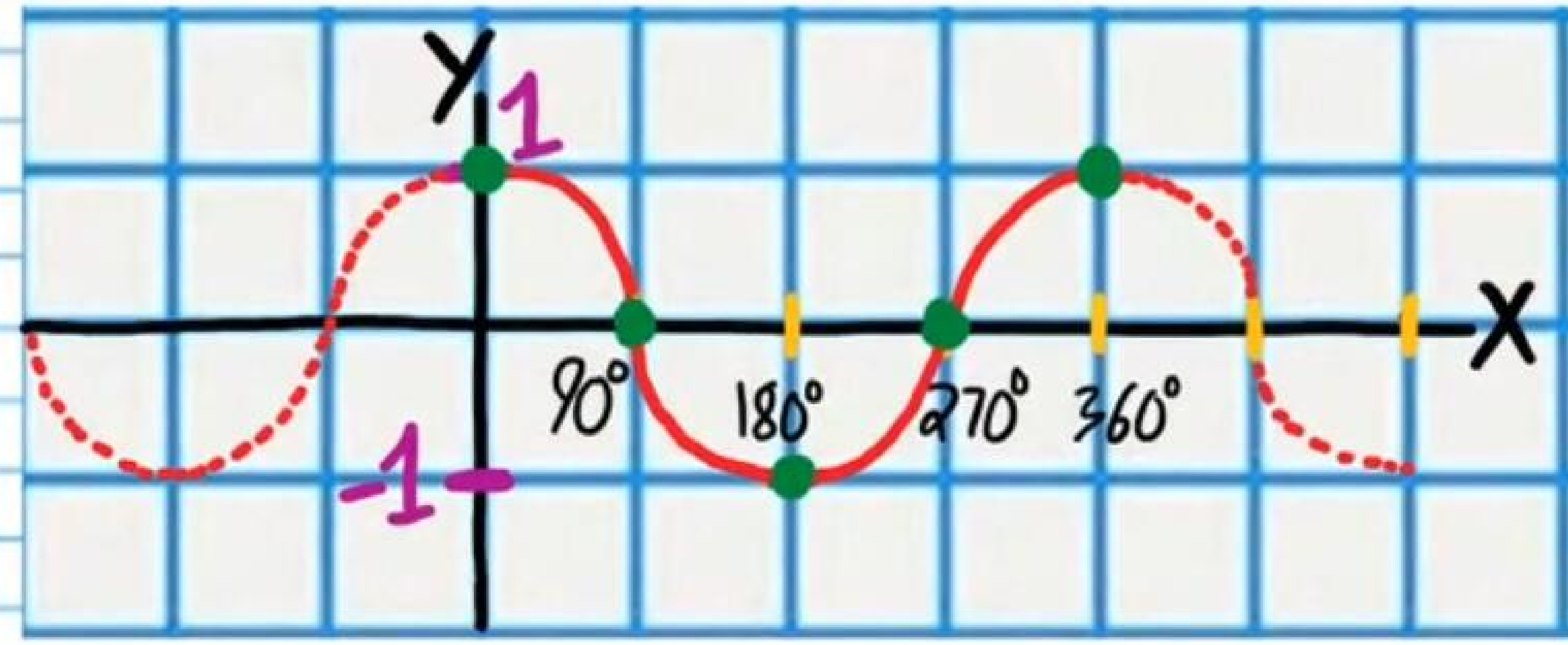
$y = \tan x$



$\boxed{\theta = 180°, 360°}$

(22) If $\cos\theta = 0$, and $0° < \theta \leq 360°$, Find the values of θ.

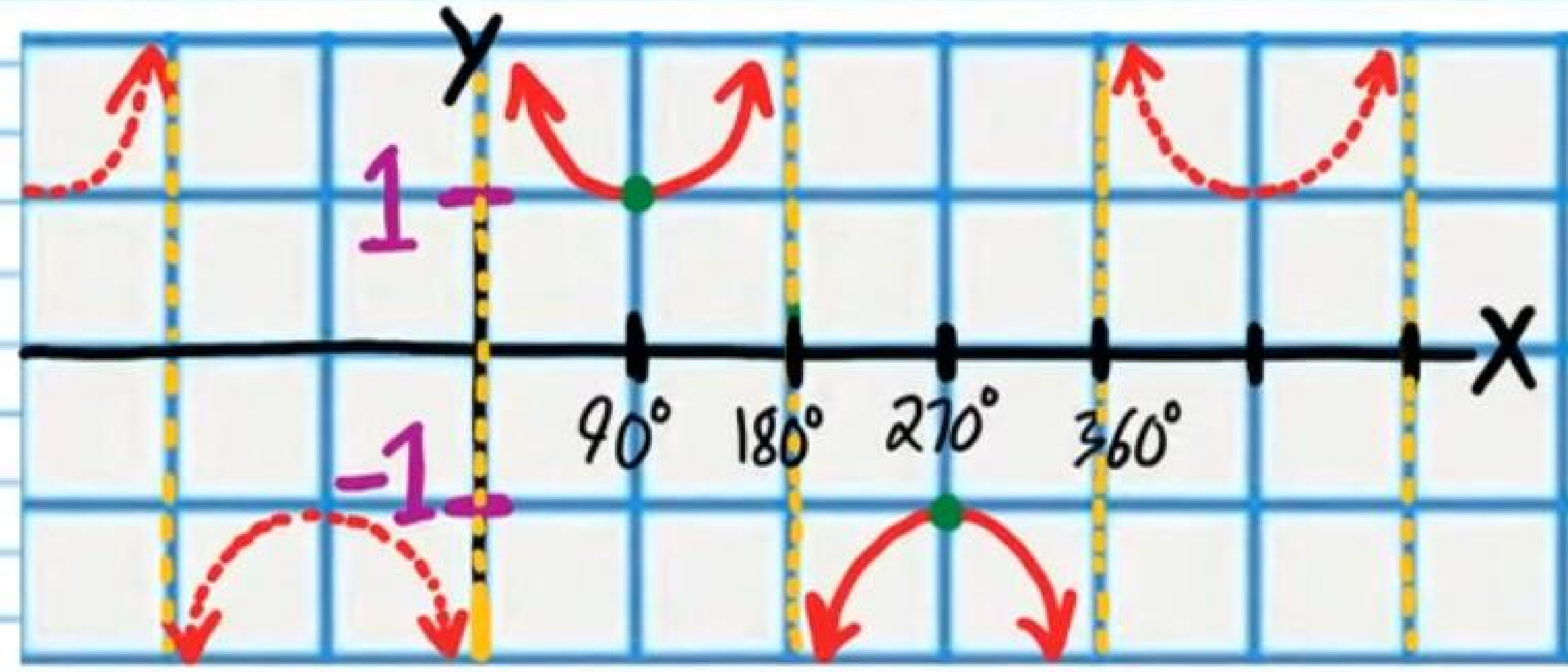
$y = \cos x$



$\boxed{\theta = 90°, 270°}$

(23) If $\csc\theta = -1$, and $0° < \theta \leq 360°$, Find the values of θ.

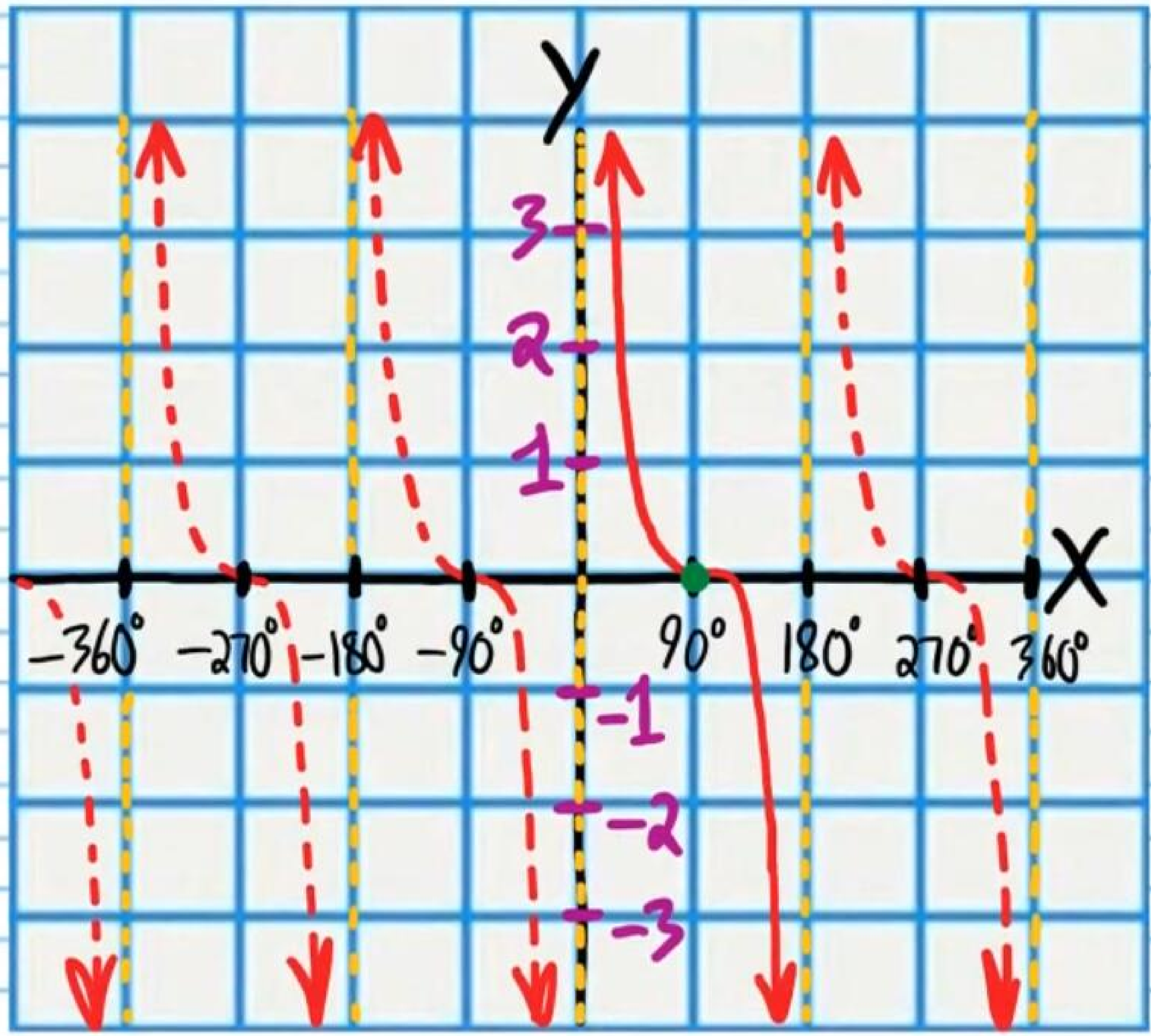
$y = \csc x$



$\boxed{\theta = 270°}$

(24) If $\cot\theta = 0$, and $0° < \theta \leq 360°$, Find the values of θ.

$y = \cot x$

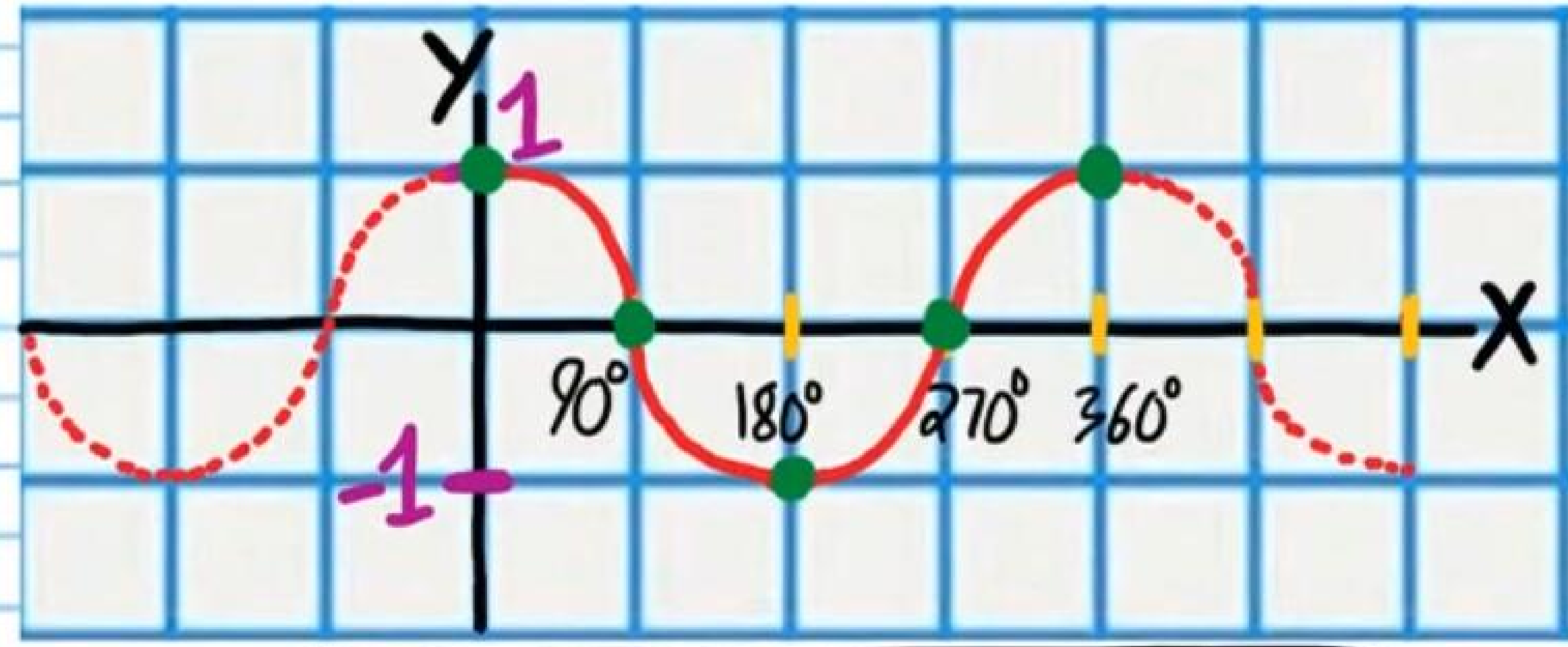


$\boxed{\theta = 90°, 270°}$

Finding the Measures of All Quadrantal Angles That Correspond to a Trigonometric Ratio

• (25) If $\cos\theta = 1$, Find all values of θ.

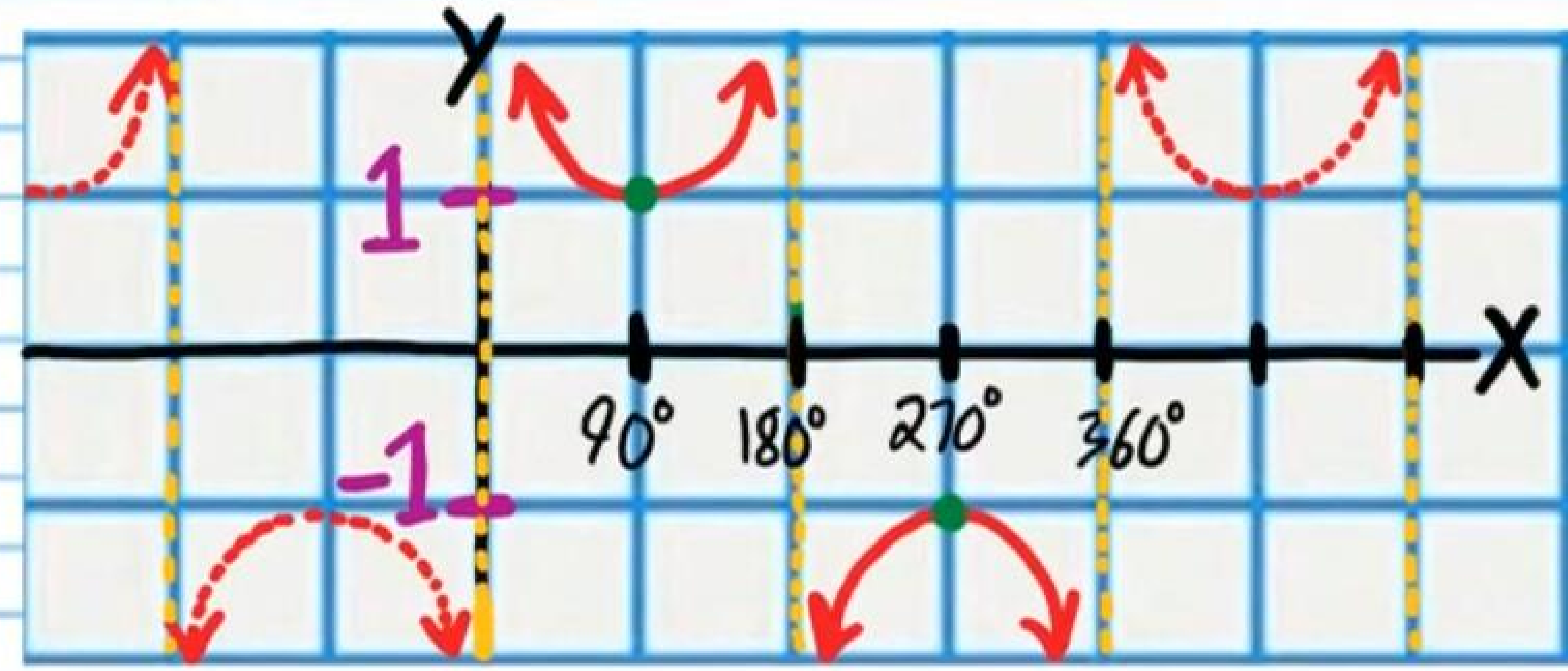
$y = \cos x$



$\boxed{\theta = 360n, \text{ n is any integer}}$

(26) If $\csc\theta = 1$, Find all values of θ.

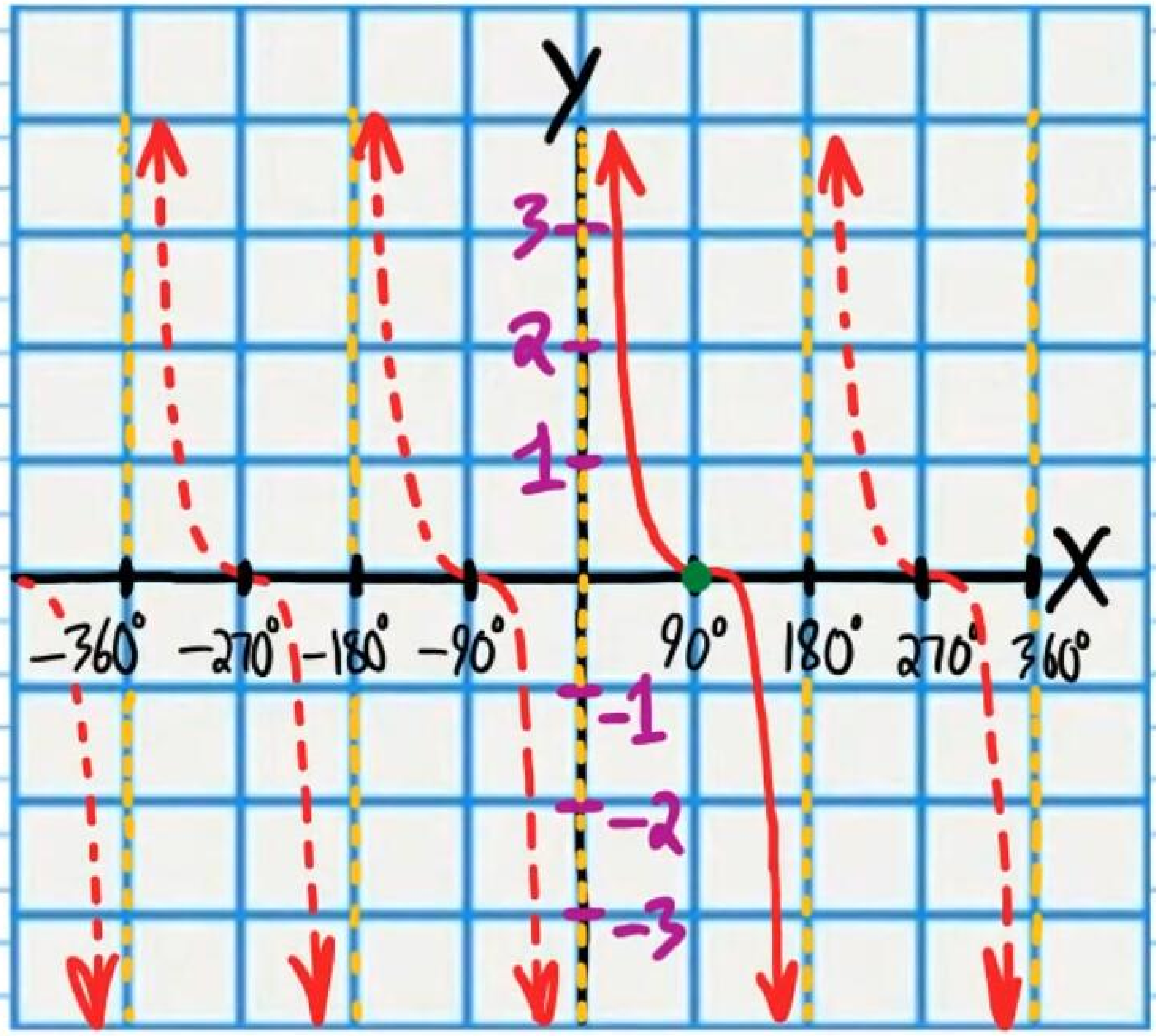
$y = \csc x$



$\boxed{\theta = 90 + 360n, \text{ n is any integer}}$

(27) If $\cot\theta = 0$, Find all values of θ.

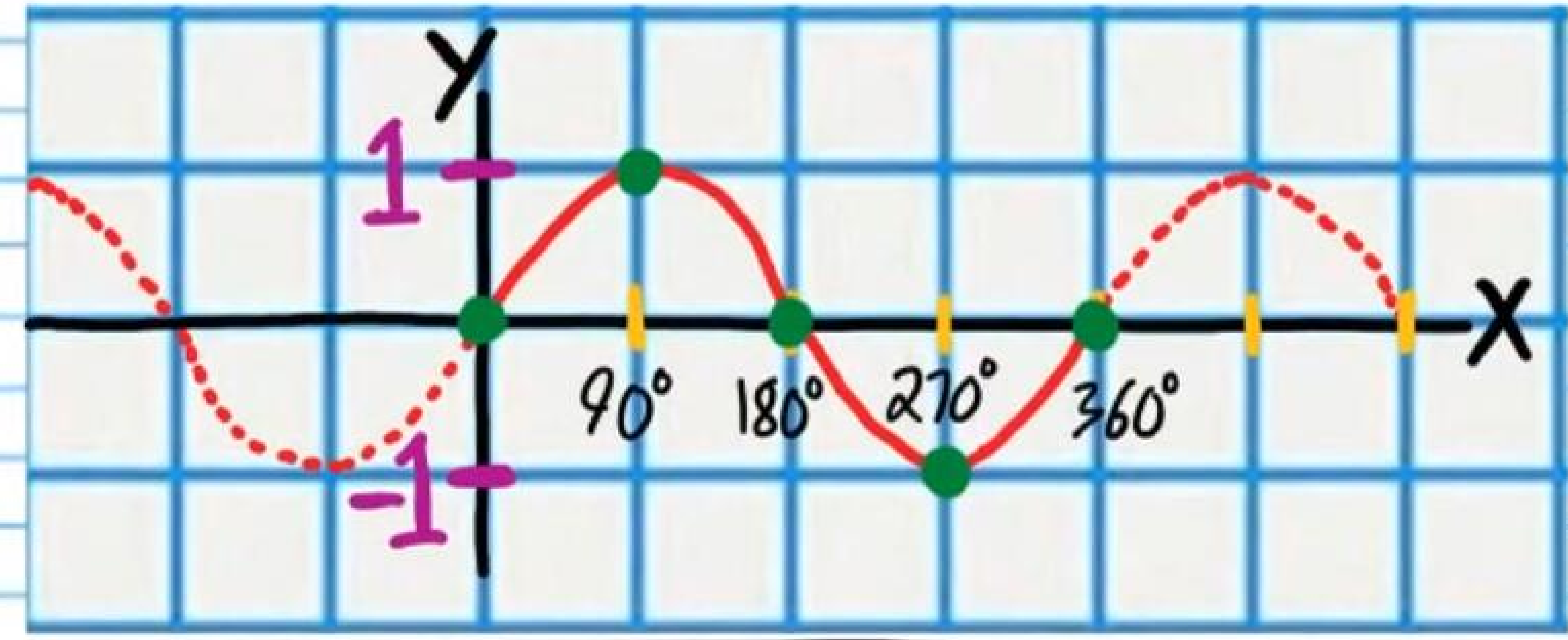
$y = \cot x$



$\theta = 90 + 180n$, n is any integer

(28) If $\sin\theta = -1$, Find all values of θ.

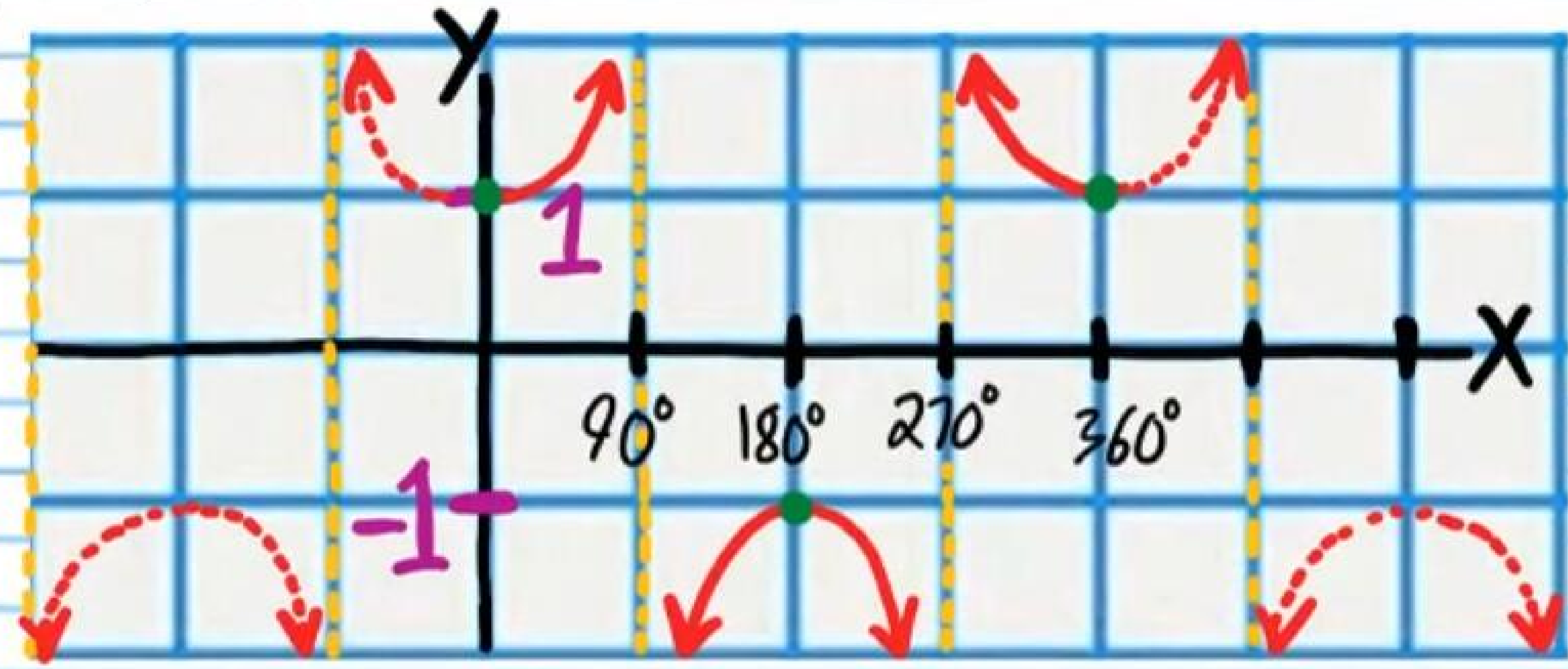
$y = \sin x$



$\theta = -90 + 360n$, n is any integer

(29) If $\sec\theta = -1$, Find all values of θ.

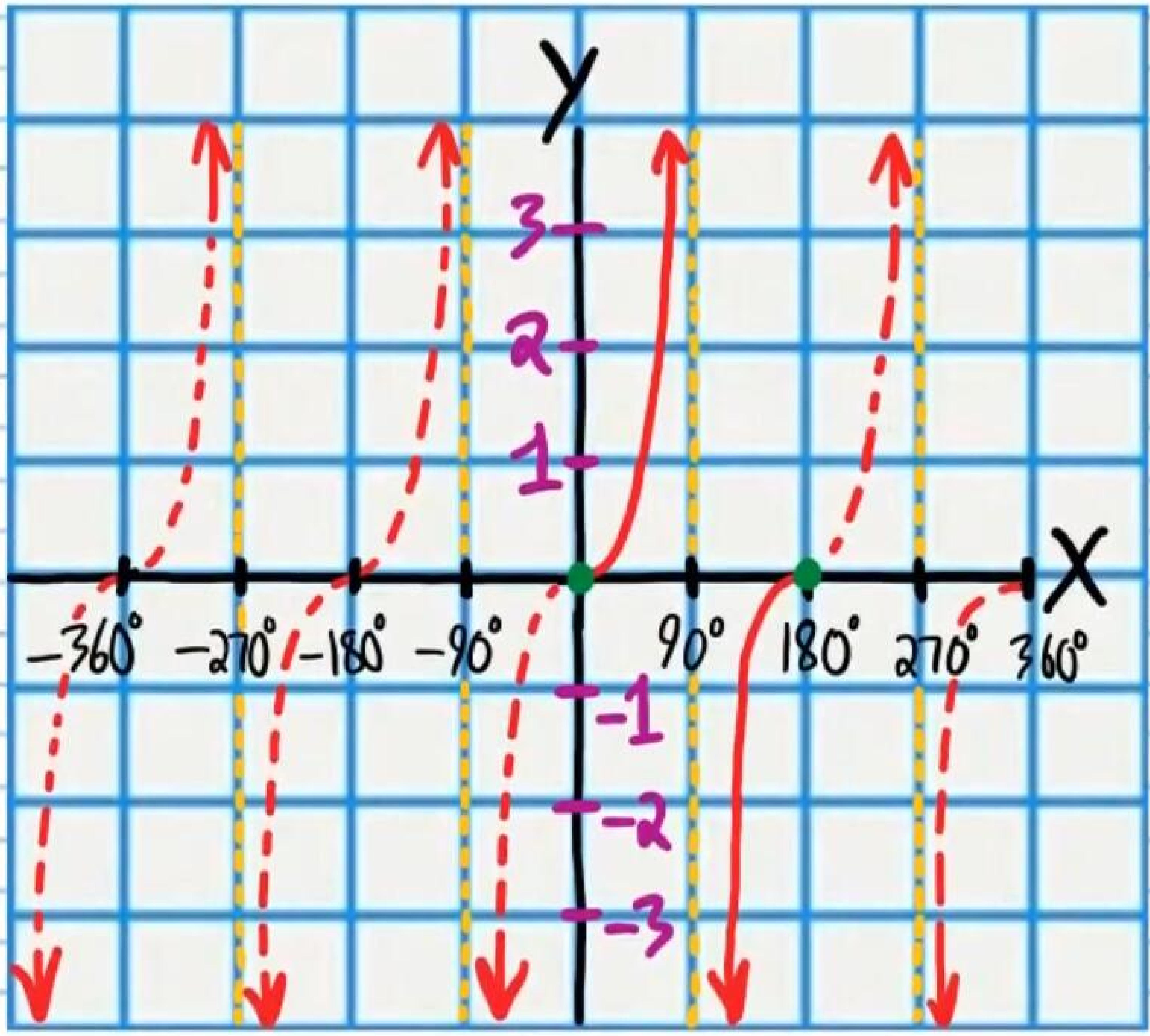
$y = \sec\theta$



$\theta = 180 + 360n$, n is any integer

30. If $\tan\theta = 0$, Find all values of θ.

$y = \tan x$



$$\boxed{\theta = 180n, \text{ n is any integer}}$$